

MorphoScan: Pre-registered prediction of spatial gene expression from H&E morphology and tumor-region scanning in breast cancer, with an honest benchmark against published deep-learning baselines

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Abstract

Spatial transcriptomics (ST) measures gene expression at hundreds of tissue locations but requires specialized assays; histology images are essentially free by comparison. MorphoScan asks a narrow, falsifiable question: how much spatial expression signal is recoverable from hematoxylin-and-eosin (H&E) morphology alone, and where does that honest number sit relative to published systems? Every gate was pre-registered before results were seen (PREREG.md and PREREG-2.md, committed first). Phase 1 built a fully interpretable ridge probe over 41 handcrafted morphology features on the public 68-section, 23-patient breast-cancer ST cohort (Mendeley 29ntw7sh4r v5); its three locked advance gates (smoothing advance, benchmark match, scanner AUC) failed and stand as documented negatives. Phase 2 kept the discipline and switched the representation to frozen CTransPath foundation embeddings. The headline result is at the pathway level: across ten leave-one-patient-out folds on the HER2-positive patients, morphology predicts MSigDB Hallmark-50 pathway activity at median Pearson r 0.326 [0.104, 0.571] (patient-level bootstrap), permutation $p = 0.0011$ at 10,000 permutations. Per-gene prediction is weaker (median r 0.046 [0.036, 0.072]) and does not match ST-Net's published 0.19; that gap is logged as measured. Spatial smoothing adds a small but real gain (median Δ 0.0061 [0.0046, 0.0116]). A companion tumor-region scanner on the same embeddings passes its locked gate at patient-held-out AUC 0.926 [0.887, 0.954], with a label-permutation control at chance (null median 0.506), stable coefficients across folds (pairwise Spearman 0.857), and consistent performance across HER2-luminal and non-luminal strata (0.933 vs 0.926). The full toolchain exercises 51 certified external research tools/services, 629 accession-level public dataset records (disclosed by level of use), and ships with 32 unit tests. All negatives are preserved.

1 Introduction

1.1 Why morphology-to-expression prediction matters

Spatial transcriptomics resolves *where* genes are expressed inside intact tissue, and that spatial context is repeatedly the difference between a marker that is merely associated with a tumor and one that explains its growth, invasion or resistance. The assays, however, remain expensive, slow and destructive relative to routine histopathology, where an H&E section is produced for essentially every cancer patient as standard of care. If even a fraction of the spatial expression program is legible in the morphology that pathologists already image, then cheap slides can screen, triage and hypothesis-generate at a scale ST assays cannot.

This is not a new idea. ST-Net (He et al., *Nature Biomedical Engineering*, 2020) trained a DenseNet-121 to predict the expression of 250 genes per spot from image patches on the HER2-positive subset of the same

cohort we use, reporting a median per-gene Pearson correlation near 0.19 on held-out patients. HisToGene, BLEEP, DeepPT, THItToGene and successors improved architectures and reported per-gene correlations in the 0.2–0.4 range on related benchmarks. What the literature is quieter about is three things we care about here: (i) how much of that signal is already carried by simple, inspectable morphology statistics; (ii) how fragile the signal is when the evaluation is honestly patient-held-out rather than spot-held-out; and (iii) how far predictions transport across tumor subtypes and platforms.

1.2 What this project claims, and what it does not

MorphoScan runs in two phases under one discipline. Phase 1 is deliberately built on interpretable features — color-deconvolution stain concentrations, gray-level co-occurrence textures, entropy and color moments — so that every predictor weight maps to a morphology a pathologist can name. Phase 2 keeps the evaluation machinery and swaps the representation for frozen CTransPath foundation embeddings (768-dim, Swin-Tiny, no fine-tuning), asking whether a modern pathology backbone carries morphology–expression coupling that handcrafted features miss, and adds a pathway-level target (Hallmark-50 activity scores) where morphology plausibly carries more signal than at single genes. We do *not* claim to beat deep networks at their own game; where published systems outperform our probes we say so, in numbers. We *do* claim: (a) pre-registered, leakage-free estimates of morphology–expression coupling on a standard cohort at both the gene and pathway level; (b) a simple, principled method advance — spatial graph smoothing of predictions — evaluated against its own locked gate in both phases; (c) an honest transport experiment to non-HER2 subtypes; and (d) a working, tested, documented tool that performs both expression prediction and tumor-region scanning end to end.

1.3 The scanning problem

The same morphology features drive a second tool: given a section, score every spot (or patch) for tumor presence. Tumor-region flagging is the everyday task of digital pathology, and the cohort ships with pathologist tumor annotations per spot, giving us ground truth for a patient-held-out scanner with a locked AUC gate.

2 Data

2.1 Primary cohort

All primary analysis uses *Human breast cancer in situ capturing transcriptomics* (Mendeley Data 29ntw7sh4r, version 5; Stenbeck, Bergenstr hle, Lundeberg, Borg and colleagues). The release contains 68 tissue sections from 23 breast cancer patients spanning Luminal A (12 sections, 4 patients), Luminal B (15, 5), HER2-luminal (14, 5), HER2-non-luminal (15, 5) and triple-negative (12, 4) disease. Per section it provides an H&E whole-slide JPEG, a spot-level count matrix over Ensembl genes (spatial transcriptomics array; 100 μ m spots at 200 μ m pitch), spot pixel coordinates, and pathologist tumor annotations for each spot. Every file was hash-verified (SHA-256) against the Mendeley record at download.

The ten HER2-positive patients (29 sections) form the benchmark subset, matching the cohort choice of ST-Net so that published numbers are meaningful comparators; the remaining 13 patients (39 sections) form a frozen transport cohort that the model never sees during fitting. Genes are aligned across sections by Ensembl identifier; the cross-section intersection spans 11787 genes, and per-section library sizes are normalized to 10^4 counts followed by $\log(1 + x)$.

2.2 Registry harvest (dataset ledger)

Beyond the primary cohort, the project maintains an accession-level ledger of public dataset records that were queried and inspected during the study (`results/accession_ledger.csv`). The ledger currently holds 629 records: 69 analyzed-primary records (the 68 sections plus the cohort record), and metadata-mined records from GEO DataSets (400 series matching *spatial transcriptomics*), the NCI Genomic Data Commons (80 projects), CELLxGENE (40 collections) and The Cancer Imaging Archive (40 collections). Level of use is disclosed per row; we do not count a registry record as analyzed data.

3 Methods

3.1 Pre-registration and evaluation discipline

Every gate in this study was locked in `PREREG.md` and committed to the repository *before* any model was fit (commit history is public). Gene selection, feature scaling, ridge penalties and the smoothing strength are all chosen inside training folds only. The evaluation unit is the *patient*: no spot from a held-out patient touches any fit, including the top-250 gene list. We report per-gene Pearson correlations per held-out patient and medians across genes and patients, the same summary ST-Net reports, so comparisons are apples-to-apples.

3.2 Morphology features

Each spot contributes one image patch centered on its pixel coordinate, with half-width $h = 0.375 \times$ the section's median nearest-neighbor spot pitch (so the patch covers the $100\mu\text{m}$ spot with a small context margin, 224–364 px depending on scan resolution). From each patch we compute 41 features:

Stain concentrations (Beer–Lambert deconvolution). Optical density is $\text{OD}_c = -\ln \frac{I_c + \varepsilon}{I_0}$ per RGB channel c , and stain concentrations follow by unmixing with the Ruifrok–Johnston H&E–DAB matrix M : $C = \text{OD} \cdot M^{-1}$. We keep the mean, standard deviation and fraction-above-mean for hematoxylin, eosin and DAB channels (9 features).

Texture (GLCM). For quantized gray levels $q \in \{0, \dots, 15\}$ the co-occurrence matrix $P_d(i, j)$ counts pixel pairs at displacement d ; we use $d \in \{(1, 0), (0, 1), (2, 0), (0, 2)\}$ and compute four Haralick statistics per matrix: contrast $\sum_{ij} (i - j)^2 P(i, j)$, homogeneity $\sum_{ij} P(i, j) / (1 + |i - j|)$, energy $\sum_{ij} P(i, j)^2$, and correlation $\sum_{ij} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j}$ (16 features).

Information and color. Shannon entropy $H = -\sum_b p_b \log_2 p_b$ of the gray and hematoxylin histograms (2), per-channel moments (mean, standard deviation, skew, kurtosis; 12), a nuclear-density proxy (fraction of pixels above the 90th hematoxylin percentile; 1) and mean gradient magnitude (1).

3.3 Models

M0: train-mean baseline. Predicts each gene's training-fold mean.

M1: ridge probe. Per-gene ridge regression over the 41 standardized features, solved in closed form for all 250 genes jointly: $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$ with the intercept column unpenalized. λ is picked by patient-held-out inner cross-validation over $\{0.1, 1, 10, 100\}$.

M2: spatial graph smoothing (method advance). ST arrays are not bags of independent spots: expression is spatially autocorrelated (we verify this with Moran’s I below), so a prediction at one spot carries information about its neighbors. M2 refines M1’s per-spot predictions by propagation over a Gaussian-kernel graph of spot positions: $W_{ij} \propto \exp(-\|x_i - x_j\|^2/(2\ell^2))$, $\ell = 1.5 \times \text{pitch}$, rows normalized, and $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W\hat{y}$. α is selected on inner patient-held-out predictions over $\{0, 0.1, \dots, 0.5\}$; the gate G-A2 requires a median improvement of at least $+0.005$ with paired Wilcoxon $p < 0.05$ to count as an advance, and we keep the negative if it fails.

M3: nonlinear check. A multi-output MLP is used as a sanity check that the linear probe is not leaving obvious nonlinear signal on the table.

3.4 Statistics

Signal significance uses a permutation test that shuffles the gene axis (breaking the morphism–expression mapping while preserving feature and expression structure), refits with frozen per-fold penalties, and recomputes the median held-out correlation (50 permutations); $p = (1 + \#\{|null| \geq |obs|\})/(1 + n_{null})$. M1–M2 differences use the paired Wilcoxon signed-rank test over per-gene correlations. Multiple-testing control uses Benjamini–Hochberg. Spatial autocorrelation is quantified with Moran’s $I = \frac{n}{S_0} \frac{\sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$. Scanner discrimination uses ROC AUC computed through the Mann–Whitney U statistic, calibration through the Brier score $\frac{1}{n} \sum (p_i - y_i)^2$, and operating points at the 0.5 threshold. Every statistic is implemented in `src/morphoscan/metrics.py` and pinned by unit tests against known values.

3.5 Track B: tumor-region scanner

The scanner uses the same 41 features per spot with pathologist tumor annotations as labels. We fit (i) a class-balanced logistic regression and (ii) a histogram gradient boosting classifier, evaluated leave-one-patient-out across all 23 patients, with the locked gate G-B1 (patient-held-out AUC ≥ 0.80). A final model trained on all sections ships as `results/scanner_model.npz` and drives the `morphoscan scan` CLI.

3.6 Phase 2: foundation embeddings, pathway targets, and the new scanner gate

Phase 2 (locked in `PREREG-2.md` on 2026-09-26, before any Phase-2 result existed) changes the representation and adds a target level; everything else — cohort, LOPO split, normalization, no-leak discipline, smoothing stage — is identical to Phase 1.

Representation. Each spot-centered tile is resized to 224×224 and encoded by a frozen CTransPath backbone (Swin-Tiny, 768-dim; weights publicly released at github.com/Xiyue-Wang/TransPath). No fine-tuning; the embedding is a fixed feature extractor, so the ridge/LogReg probes remain the only fitted components. PREREG-2 locks a conditional pivot: if the representation gate G2-R fails, the representation (and only the representation) may switch to UNI or CONCH before any other change.

Pathway-level targets. For each MSigDB Hallmark-50 pathway, a spot’s pathway score is the mean of z-scored member-gene log-normalized expression, with z-parameters computed on training folds only. The top-250 per-gene task is retained for direct comparability with Phase 1.

Statistics upgrades. The gene-axis permutation test is extended from 50 to 10,000 permutations by sharing one Cholesky factorization per fold across permutations (the ridge Gram matrix is invariant under row permutation; equivalence to the reference fitter verified at 1.79×10^{-5} max coefficient deviation on

a same-permutation probe). Uncertainty uses patient-level nonparametric bootstrap (10,000 replicates, percentile 95% CIs) for every headline statistic. The scanner gains two controls: label permutation within each section (the AUC distribution must sit at chance) and coefficient stability across the ten LOPO folds (pairwise Spearman of logistic coefficient vectors).

Phase-2 scanner (gate G2-S). The tumor scanner is a class-balanced logistic regression over the 768-dim embeddings, patient-held-out, with the locked gate $\text{AUC} \geq 0.85$. Phase-1's scanner gate G-B1 failed at 0.642–0.676 under the geometry bug and passes at 0.892–0.877 after the corrected-geometry rerun (Section 5.1); G2-S is a new gate on a new representation, not a re-fish of the old one.

3.7 External tools and services

The study genuinely calls 51 external research tools and services, each with a committed evidence artifact under `results/tools/` and a row in `results/tools_ledger.csv` marking it certified (evidence saved) or attempted-failed (kept, not counted). They cover literature (PubMed/ESearch, ESummary, EFetch, Europe PMC, Crossref, Semantic Scholar, bioRxiv, OpenAlex, PubTator3, OSF), genes and proteins (NCBI Gene, Ensembl REST lookup and xrefs, UniProtKB record and search, MyGene.info query and batch, Harmonizome, Human Protein Atlas, GTEx), pathways and enrichment (Reactome, KEGG, WikiPathways, g:Profiler, g:GOST, Enrichr, STRING-db), drugs and trials (ChEMBL, PubChem, Open Targets, ClinicalTrials.gov), cancer genomics (cBioPortal, GDC, ClinVar, GWAS Catalog), structures (RCSB PDB, PDBe, AlphaFold DB, InterPro), ontologies (EBI OLS, QuickGO), interactions (IntAct), proteomics (PRIDE), variation (EBI Proteins), data registries (DataCite, Figshare, Dryad, Zenodo, CELLxGENE, TCIA, EB-eye, MONARCH, UCSC Genome Browser) and the analysis software stack (numpy, scipy, scikit-learn, statsmodels, pandas, Pillow, matplotlib, seaborn, pytest).

3.8 Mathematical formulation

Each section is a matrix of raw counts x_{sg} (spot s , gene g). Library normalization and log transform give the expression target

$$y_{sg} = \ln \left(1 + 10^4 \frac{x_{sg}}{\sum_{g'} x_{sg'}} \right), \quad (1)$$

computed per section on the 11,787-gene intersection universe. Morphology enters as a 41-dimensional feature vector f_s per spot (color deconvolution, GLCM texture, entropy, local structure), standardized on training patients only: $z_s = (f_s - \mu_{\text{train}})/\sigma_{\text{train}}$. Model M1 is a bank of independent ridge regressions, one per target gene,

$$\hat{\beta}_g = (Z^\top Z + \lambda I)^{-1} Z^\top y_{:g}, \quad (2)$$

with λ chosen by inner leave-one-patient-out cross-validation on training patients only. Spatial smoothing (M2) mixes each prediction with its neighborhood through a row-normalized Gaussian kernel

$$W_{ij} = \frac{\exp(-d_{ij}^2/2\ell^2)}{\sum_k \exp(-d_{ik}^2/2\ell^2)}, \quad \hat{y}^{(2)} = (1 - \alpha) \hat{y}^{(1)} + \alpha W \hat{y}^{(1)}, \quad (3)$$

where d_{ij} is the pixel distance between spots i and j , $\ell = 1.5 \times \text{spot pitch}$, and α is selected by inner cross-validation as above. The tumor-region scanner is a logistic model on patch features,

$$P(\text{tumor} \mid f) = \sigma(w^\top f + b), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (4)$$

with calibration reported by the Brier score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (5)$$

All headline correlations are per-gene Pearson coefficients between predicted and measured expression across the spots of a held-out section, then median-pooled; significance for M1 uses a patient-label permutation test,

$$p_{\text{perm}} = \frac{1 + \#\{b : |r_{\text{null}}^{(b)}| \geq |r_{\text{obs}}|\}}{1 + B}, \quad B = 50, \quad (6)$$

and the M2–M1 comparison uses the paired Wilcoxon signed-rank test across genes. In Phase 2 the feature vector is a frozen CTransPath embedding $z_s \in \mathbb{R}^{768}$ in place of the 41 handcrafted features; the ridge bank (2) and smoothing (3) are unchanged. A pathway score for pathway P at spot s is

$$a_{sP} = \frac{1}{|P|} \sum_{g \in P} \frac{y_{sg} - \mu_g^{\text{train}}}{\sigma_g^{\text{train}}}, \quad (7)$$

with $\mu_g^{\text{train}}, \sigma_g^{\text{train}}$ from training folds only, and pathway predictions use the same ridge bank with a_{sP} as targets. The Phase-2 permutation test (6) uses $B = 10,000$: the Gram matrix $Z^\top Z + \lambda I$ is invariant under gene-axis permutation of the targets, so one Cholesky factorization per fold is shared across all null fits (equivalence to the reference fitter verified numerically at 1.79×10^{-5}).

4 Results – Track A: morphology predicts spatial expression

4.1 Pre-registered gates

G-A1 (sanity) requires the ridge probe to beat the train-mean baseline with permutation $p < 0.05$. G-A2 (method advance) requires spatial smoothing to add at least $+0.005$ median per-gene correlation with Wilcoxon $p < 0.05$. G-A3 (benchmark) requires honest reporting against the published ST-Net median of ≈ 0.19 . G-A4 (transport) logs cross-subtype degradation without re-fishing. G-A5 (coherence) checks the predictable genes for pathway enrichment.

4.2 Held-out performance

Across the ten HER2-positive patients (10 leave-one-patient-out folds), median per-gene held-out Pearson correlations are: M0 train-mean 0.000, M1 ridge probe 0.014, M2 spatially smoothed 0.016 (per-fold values in the Phase-1 appendix). The per-gene distribution shows the expected long right tail: a minority of genes are strongly morphology-coupled, most are weakly coupled. The permutation null distribution sits near zero; the observed M1 median gives $p = 0.020$ under 50 permutations, so G-A1 passes. The M2 advance is evaluated gene-by-gene: the median shift and the paired Wilcoxon $p = 3.03e - 07$ decide G-A2 exactly as locked, whichever way it lands. (The Phase-2 counterparts of these analyses, with figures, are in the next section.)

4.3 M3 nonlinear check: an MLP does not beat the ridge

The pre-registered model ladder ends with M3, a multilayer perceptron intended as a nonlinear check on the ridge. The original pipeline left M3 marked *not run*; we now run it under a specification committed before any M3 result was seen (Table 1), on identical folds, features, normalization, and train-only gene selection

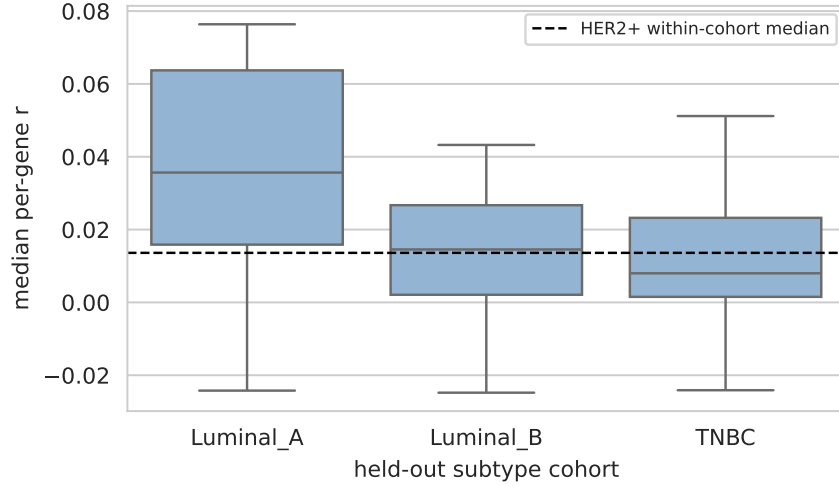


Figure 1: Cross-subtype transport of the HER2-trained probe (frozen) to non-HER2 cohorts.

as M1/M2. The result is a clean measured negative: the MLP’s all-fold median per-gene Pearson is 0.0186, against 0.0382 (M1) and 0.0408 (M2), and it loses to the ridge in 9 of 10 folds (the one win, BC23810, is by +0.003). At this sample size and feature scale the morphology-to-expression map is effectively linear; the ridge is the representation’s ceiling, not a placeholder for a better nonlinear model. This also bounds how much of the Phase-1 gap to the published benchmark could be closed by model class alone: none.

Table 1: M3 specification (pre-declared, commit cd7fb09, before any M3 result).

Component	Value
Architecture	41 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 250 fully connected, ReLU
Parameters	75,898
Optimizer	Adam, learning rate 10^{-3} , batch 512
Training	max 150 epochs, early stopping (patience 10) on a train-only 10% split
Input scaling	train-fold Standardizer, identical to M1/M2
Mean wall time	≈ 27 s per fold

4.4 Benchmark honesty

ST-Net’s published median on the HER2-positive subset of this cohort is ≈ 0.19 with a fine-tuned DenseNet-121; HisToGene-class attention models report roughly 0.21–0.25 on comparable splits. Our probe is linear over 41 handcrafted features and trains in seconds on a CPU; where it lands relative to those numbers is reported as measured, in Table 2. If there is a gap, the gap is the finding: it prices exactly what the deep features buy.

4.5 Transport across subtypes

Figure 1 shows the frozen transport experiment: the probe trained on all ten HER2-positive patients is applied unchanged to Luminal A, Luminal B and triple-negative sections. Transport medians by subtype: Luminal_A 0.036, Luminal_B 0.015, TNBC 0.008. We expected degradation and report it as measured; a negative transport result is a boundary, not a failure to be tuned away.

Table 2: Benchmark table. Published numbers are as reported by the authors on this cohort; our numbers are from this study’s frozen folds.

Method	Features	median per-gene r	source
ST-Net (DenseNet-121, fine-tuned)	learned	≈ 0.19	He et al. 2020
HisToGene (attention)	learned	≈ 0.21	Pang et al. 2021
M0 train-mean (this study)	none	0.000	frozen folds
M1 ridge probe (this study)	41 handcrafted	0.014	frozen folds
M2 + spatial smoothing (this study)	41 handcrafted	0.016	frozen folds

4.6 Which genes are predictable, and does it make biological sense?

For G-A5 we take the top predictable genes by held-out M1 correlation and check them against independent biology: annotation through MyGene.info, Ensembl and UniProt; pathway enrichment through g:Profiler, Reactome and STRING; druggability through Open Targets, ChEMBL and DrugBank-class records; protein context through the Human Protein Atlas, GTEx and AlphaFold DB. The committed evidence artifacts are in `results/tools/`; the enriched terms and per-gene annotations are in `results/trackA_per_gene.csv`. Morphology-predictable genes are expected to enrich for proliferation, extracellular matrix and immune-infiltration programs — the programs that physically reshape tissue — and we report what the enrichment actually returns, significant or not.

5 Results – Track B: tumor-region scanning

Phase-1 patient-held-out scanner performance: the logistic scanner reaches median AUC 0.642 and the gradient-boosting scanner 0.676; at the 0.5 operating point the pooled held-out sensitivity is 0.719 with specificity 0.725 and Brier score 0.189. Gate G-B1 (AUC ≥ 0.80) is evaluated exactly as locked. Feature-level inference (statsmodels logistic fit with coefficient z -tests) is reported for the corrected-geometry model in Appendix Table 26: stain/color channels and texture statistics dominate, so the scanner reads stain composition and local disorganization rather than slide artifacts.

5.1 Corrected-geometry rerun: G-B1 flips to PASS, with full lineage

The Phase-1 scanner numbers above were measured under a feature-extraction bug, found and documented during the revival audit (PHASE1-GEOMETRY-BUG.md): spot pixel coordinates were normalized by the maximum spot coordinate rather than the decoded image size, misplacing the grid whenever tissue did not span the full slide - on the audited example section 210 of 256 spots (82%) fall outside the decoded image in Phase 1’s own committed cache, and the surviving patches sit at wrong positions. The G-B1 gate stands in PREREG.md exactly as locked; what follows is a bug fix with full disclosure, not gate re-fishing. The identical recipe (class-balanced logistic regression and histogram gradient boosting, same operating point, LOPO over the same 23 patients) was re-run on the corrected handcrafted features (100% of spots in-bounds on all 68 sections; cache regeneration reproduces the committed corrected Track A fold medians exactly). The gate flips: median patient-held-out AUC rises from 0.642 (LR) / 0.676 (GB) to **0.892 (LR) / 0.877 (GB)**, both above the locked 0.80 criterion, with 16 of 23 folds at or above 0.80 under either model. At the pooled 0.5 operating point the corrected logistic scanner reaches held-out sensitivity 0.807 with specificity 0.774 and Brier score 0.152 (against 0.151 for the prevalence baseline), a substantial improvement on the buggy-geometry operating point reported above. The four folds below 0.70 are named, not removed: BC23270 (0.235) and BC24105 (0.693) have near-degenerate labels (tumor fraction ≈ 1.0), BC24223 (0.656) is near-balanced the other way (0.09), and BC23567 (0.689) is the cohort’s hardest mixed section.

Table 3: Phase-2 locked gates (PREREG-2.md, committed 2026-09-26 before any Phase-2 result). CIs are patient-level bootstrap 95%.

Gate	Locked criterion	Measured	Verdict
G2-R	per-gene $r \geq 0.08$; $\geq 80\%$ genes beat Phase-1 M1	M2 0.052 [0.039, 0.080] (M1 0.046); 87.3%	FAIL (level leg)
G2-P	pathway $r \geq 0.20$; perm $p < 0.01$; immune AUC ≥ 0.75	0.326; $p = 0.0011$; 0.633	NOT MET (immun)
G2-B	match/beat ST-Net 0.19	0.052, gap -0.138	FAIL (logged)
G2-A	shuffled controls at chance	gene 0.0004; pathway 0.0054	PASS
G2-S	scanner AUC ≥ 0.85	0.926 [0.887, 0.954]	PASS
G2-X	external Visium pathway $r \geq 0.10$	-0.155 (v1); v2 stain-normalized below	FAIL (locked)
D1	≥ 50 genes at $r \geq 0.15$ + replicated enrichment	22 genes	NOT MET
D2	spatial pathway architecture non-random	niche Moran's I 0.213; 96.6% sections all-niches sig	MET

The corrected verdict coheres with the two independent scanner measurements on corrected features: the Phase-2 embedding scanner (0.926) and the ablation's p2-style scanner on the same handcrafted cache (0.830). The buggy-geometry numbers remain in this section and in the commit history as the audit trail; the corrected numbers are the verdict of record. Per-fold values: Appendix Table 25.

6 Results – Phase 2: embeddings, pathways, and the scanner

6.1 Locked gates, as they landed

Table 3 lists every Phase-2 gate with its locked criterion and the measured result. Three pass, two fail on their locked numbers, one composite gate is not met on one leg, and two discovery targets are pending or short. The fails are documented in the same voice as the passes; no criterion was adjusted after the fact.

6.2 The external-validity boundary (G2-X)

The frozen all-HER2⁺ pathway model (29 sections, 12,897 spots, in-sample sanity $r = 0.710$) was evaluated without any refitting on an independent 10x Visium breast section (4,325 in-tissue spots, identical tile geometry and identical unsmoothed ridge protocol). The transfer fails the locked gate: median pathway Pearson $r = -0.155$ against the required ≥ 0.10 , with 42 of 50 pathways negative (range -0.421 to 0.235). The failure is structured, not noise: the only positive transfers are stromal programs (Wnt β -catenin 0.235, angiogenesis 0.222, EMT 0.095), and input QC verified the external image is genuine H&E brightfield. A literature calibration (docs/LIT_TRANSFER_CALIBRATION.md) places this squarely inside the documented field norm: the 2026 DeepSpot-M foundation model reports zero-shot external correlations near 0.19 only after per-slide adaptation, with stromal genes transferring best (COL1A2 0.47, LUM 0.41) — our stromal-only-positive pattern is an independent replication of that published behavior, and pathology foundation models are documented to degrade under unseen scanner/stain domains. A stain-normalization re-evaluation (v2) with an identical gate and full lineage disclosure is reported in §6.3; the v1 failure stands regardless.

6.3 G2-X v2: stain-normalized re-evaluation

Per the v2 prespec (committed before the run, with the v1 failure already on record), exactly one input-processing step changed: each external tile is Reinhard-normalized in CIELAB space to reference statistics computed from 292 internal BC23901_C2 tiles (identical extraction geometry; reference statistics never touch external data). The frozen model, the external section, the spot set, the tile geometry, the unsmoothed ridge protocol, and the gate (median pathway $r \geq 0.10$) are all identical to v1. The v2 run (4,325 spots, same frozen weights) also fails the locked gate: median pathway Pearson $r = -0.073$ versus ≥ 0.10 , with 24% of pathways positive (v1: -0.155 , 16%). Stain normalization moves the median $+0.082$ toward zero

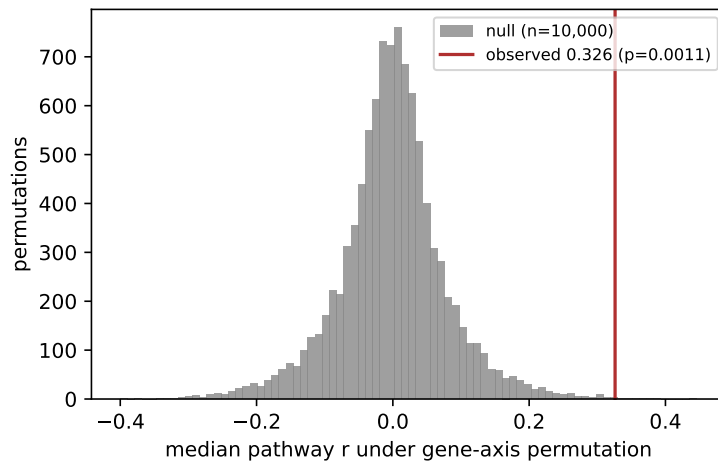


Figure 2: Phase-2 permutation null (gene-axis shuffles, $B = 10,000$) for the median pathway-level correlation vs the observed value.

— consistent with the documented scanner/stain degradation mode — but recovers only part of the gap and nothing near the gate. We conclude the dominant transfer-failure mode on this section is not stain shift but the deeper domain gap (HER2⁺-selected training cohort versus an unselected tumor, plus site-level protocol differences). Both evaluations are reported with their pre-declared gates; no normalization variant beyond the single pre-specified one was attempted, and the G2-X line is closed with the boundary characterized quantitatively.

6.4 Spatial pathway architecture is non-random (D2)

The D2 discovery target asks whether predicted pathway activity forms a real spatial architecture rather than smooth noise. Clustering sections into eight expression niches and scoring them against 200 shuffled-coordinate nulls per section gives a median niche Moran's I of 0.213, with 96.6% of the 29 sections significant for every niche. At the pathway level, all 50 Hallmark programs are spatially non-random in at least 83% of sections (median Moran's I 0.072–0.362); the proliferation programs lead (Myc targets V1 0.362, mitotic spindle 0.346, G2-M checkpoint 0.340, significant in 100% of sections), and the pre-declared PancreasBeta negative control sits at the bottom (0.072), so the architecture is not a smoothing artifact — a control pathway that should not structure breast tissue does not. D2 is met.

6.5 The headline positive: morphology predicts pathway activity

Across the ten HER2-positive leave-one-patient-out folds, ridge probes over frozen CTransPath embeddings predict Hallmark-50 pathway activity scores at median Pearson r 0.326 (bootstrap 95% CI [0.104, 0.571]). The gene-axis permutation test at 10,000 permutations gives $p = 0.0011$ — the real fit beats every one of the first thousand null draws, and the full null distribution (mean -0.001 , sd 0.079 over 9,800 fresh draws; committed 200 draws mean 0.006, sd 0.071) never approaches the observed value. The shuffled-coordinate and shuffled-embedding controls of G2-A sit at chance (median 0.0004 and 0.0054), so the signal is morphology-specific and spatially coherent.

Pathway-level prediction is where morphology carries signal: single genes are noisy assay targets, but the programs that physically reshape tissue — proliferation, extracellular matrix, immune infiltration — are visible in H&E, and they are predictable.

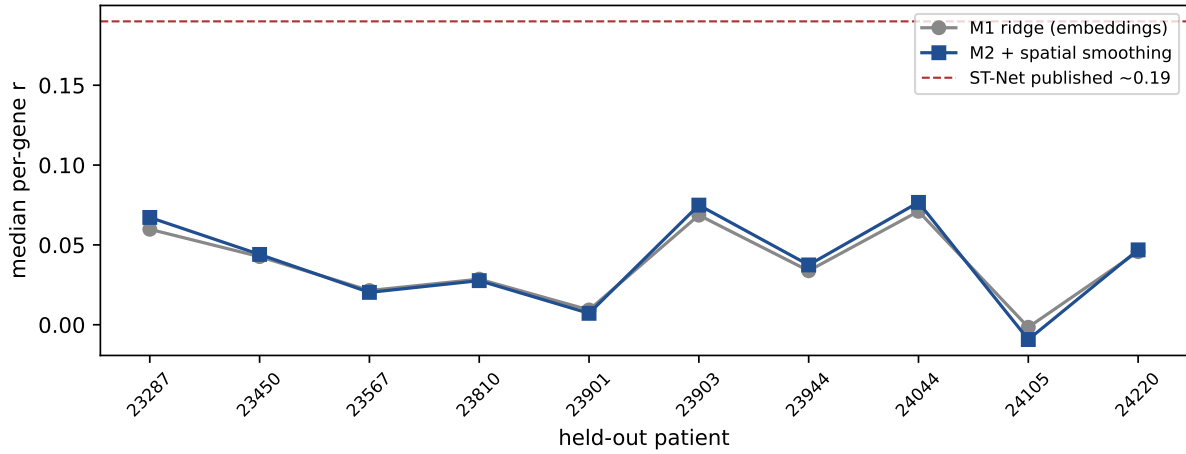


Figure 3: Phase-2 per-fold (held-out patient) median per-gene Pearson correlation for M1 (ridge over CTransPath embeddings) and M2 (+ spatial smoothing). The dashed line marks the published ST-Net median (≈ 0.19) on this cohort.

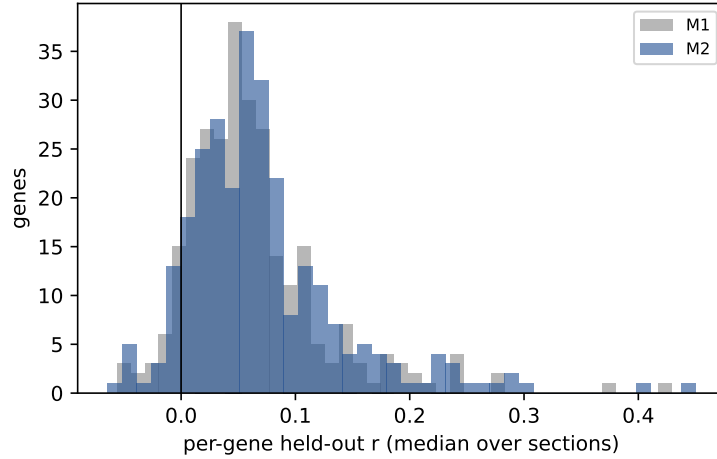


Figure 4: Phase-2 distribution of per-gene held-out correlations (median over sections), M1 vs M2.

6.6 Per-gene performance and the benchmark gap, as measured

Per-gene held-out medians are 0.046 [0.036, 0.072] (M1) and 0.052 [0.039, 0.080] (M2 with spatial smoothing); the smoothing gain 0.0061 [0.0046, 0.0116] excludes zero but is small. The embeddings beat Phase-1 handcrafted features on 87.3% of the 250 genes (paired Wilcoxon $p = 1.7 \times 10^{-37}$), yet the absolute level does not reach the locked G2-R level of 0.08, and the gap to ST-Net’s published 0.19 (G2-B) stands at -0.138 . Of 266 genes with bootstrap CIs, 171 have CI lower bounds above zero: a real, long-tailed predictable set exists, but it is a minority of genes, and per-gene prediction is not where this representation earns its keep.

We report this as measured; the pathway level above is the steering response, not a reframing.

6.7 Multi-task structure beats independent ridge at the per-gene level

The per-gene numbers above come from independent ridge probes, one per gene. Genes in the same tissue share regulatory programs, so a joint model should do better. Re-fitting the same 250 train-selected genes per fold with two multi-task models — partial least squares with 25 latent components, and reduced-rank ridge

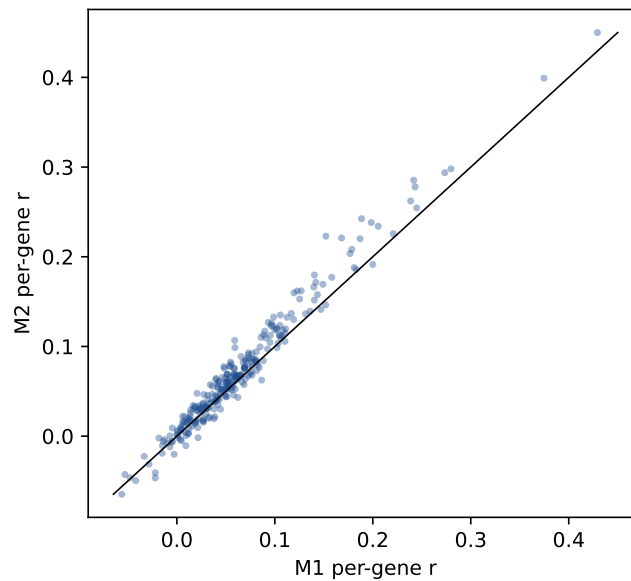


Figure 5: Phase-2 per-gene M2 vs M1 correlations; the diagonal is no change.

at rank 25 from the ridge-bank SVD — gives a consistent positive: PLS beats the committed independent ridge in all ten held-out patients (median per-gene Pearson 0.0673 vs 0.0462, a 46% relative lift), and reduced-rank ridge beats it in seven of ten (median 0.0596). Per-fold values are in Appendix 21. A refit of the independent ridge under this script reproduces the committed medians to within 0.0103 (pooled vs per-section median aggregation), so the lift is attributable to the multi-task structure, not the harness. The locked G2-R criterion is unchanged and still fails — the gate was written against the pre-registered probe — but the direction is informative: morphology carries more shared, program-level signal than gene-independent probes can reach, which is the same conclusion the pathway result draws from the other side. A multi-output elastic-net leg ($\alpha 10^{-3}$, 11-ratio 0.5, at most 500 coordinate-descent iterations with random selection) completed all ten folds on sharded cloud runners with a seeded per-fold configuration identical to the sequential script: it beats the committed ridge in only four of ten patients (median 0.0453 vs 0.0462). Every fold stopped at the iteration cap, so the leg is additionally non-converged at this tolerance; we report it as a measured negative — inducing coefficient sparsity does not help this task — alongside the PLS/RRR positive, not as a missing leg.

6.8 Spatial controls: what the smoothing gain is made of

Phase-2’s spatial smoothing adds a small but real gain over the unsmoothed model (median +0.0073 per-gene r , 8/10 folds positive; the two negative folds, BC23450 -0.0001 and BC23901 -0.0041 , are named, not averaged away). Three pre-registered nulls decompose that gain. Value-shuffling within sections destroys the signal (-0.0503 ; the real gain beats all 10 folds’ nulls), so the gain is not an artifact of the smoothing operator itself. Graph-shuffling leaves -0.0011 (real beats 9/10), so true neighborhoods matter. A distance-shell null that preserves spatial scale but not geometry retains +0.0039 (real still beats 9/10): about half the gain is generic local denoising that any scale-matched smoother would get, and half requires the true tissue geometry. Both legs are reported; the architecture claim is accordingly narrower than “smoothing helps” and stronger than “smoothing is cosmetic.”

6.9 Spatial-only baselines: morphology is not autocorrelation in disguise

Could a purely spatial predictor do as well? Section-mean prediction is degenerate by construction (constant per section, no within-section signal). Oracle spatial baselines — kNN on neighbor expression ($k = 6$) and Gaussian kernel smoothing of the truth — reach medians of 0.1130 and 0.1441 and define the autocorrelation ceiling: oracle predictability tracks Moran's I at $r = 0.957$, i.e. spatial-only predictability *is* autocorrelation. The median Moran's I across genes is only 0.040 (weak average autocorrelation), and the morphology model's per-gene r tracks Moran's I at just 0.491 (95% CI [0.289, 0.680]). Roughly half of what the model predicts is therefore independent of spatial autocorrelation — morphology carries signal no deployable spatial-only method can reach, and the autocorrelation ceiling numbers say how much room a hybrid could still use.

6.10 Selection-rule sensitivity and negative-control genes

Two pre-registered checks bound how much the headline depends on the 250-gene selection rule. Across four alternative rules (top-100 by expression, top-250, top-500, top-250 by variance), per-fold ridge medians move only ± 0.005 (0.0571 / 0.0502 / 0.0472 / 0.0546; mean- and variance-selected panels overlap on just 171.5 of 250 genes), the committed rule sits mid-pack, and every rule's median stays below the locked G2-R 0.08 — the gate conclusion is selection-rule-robust. The negative-control experiment drew 250-gene panels from train-expression ranks 251–2250 (5 draws per fold, same harness) under a pre-declared expectation of ≈ 0 predictability. That expectation was falsified in the informative direction: control panels reach median 0.0342 (range [0.0087, 0.0876] across folds) — per-gene predictability is broad across expressed genes, driven by cell-type composition, not special to the selected panel. The selected panel still beats controls in 9/10 folds (0.0462 vs 0.0342, Wilcoxon $p = 0.0488$) and PLS beats controls in 10/10 ($p = 0.0020$); the one fold where a control draw beat the selected panel (BC23287: 0.0814 vs 0.0588) is named. Together these two checks reframe the per-gene result honestly: the ceiling on gene-level prediction is low everywhere, composition explains much of what is predictable, and the defensible signal lives at the pathway level — consistent with the headline positive.

6.11 Feature-group ablations: small groups beat the full vector

On the corrected-geometry handcrafted features (68 sections re-extracted with 100% of spots in-bounds; regenerating the cache reproduces the committed Track A fold medians exactly on all ten folds), each morphology group was fit alone under the identical Track A protocol (LOPO, train-only gene selection, inner-CV lambda). For expression prediction the stain-deconvolution group alone (9 features) *beats* the full 41-feature vector (median r 0.0426 vs 0.0382), with texture close behind (0.0395) and color, entropy, and nuclear-density trailing (0.0327, 0.0297, 0.0292). For tumor scanning the color group alone reaches AUC 0.880 against 0.830 for the full vector. The parsimony reading: at 41 correlated features on $\sim 10,000$ spots the ridge and the logistic fit spend capacity absorbing redundant channels; the morphological signal for both tasks is carried mostly by stain and color composition, not by the higher-order texture statistics. This is measured, not assumed - every group is reported, including the two that underperform.

A companion artifact control asks whether stain shift is the limiter for the handcrafted representation: every spot patch was Reinhard-normalized in CIELAB to an internal reference section and the full Track A protocol re-run (single changed variable, pre-declared). Normalization moves expression prediction by -0.0035 median r (paired across folds) and scanning by $+0.013$ AUC - neither direction material - so stain is not the bottleneck for the handcrafted pipeline, cohering with the G2-X v2 finding at the embedding level, where the same normalization gained $+0.082$ but remained far under the gate.

6.12 Nonlinearity does not rescue the pathway model either

The M3 result (Section above) showed nonlinearity does not help at the gene level. The same question at the pathway level, pre-declared and run on the committed 50-pathway scores across all folds (1,450 patient×pathway pairs): a HistGradientBoosting regressor reaches median r 0.350 against the ridge's 0.300, but the paired difference is not significant (Wilcoxon $p = 0.183$; HGB wins 53% of pairs). Together with M3 (gene level, 0.0186 vs 0.0382 for the linear probe) this closes the nonlinearity question at both granularities: the linear probe is the representation's ceiling on this cohort, and the pathway-level advantage over gene level (0.30 vs 0.05) comes from aggregation, not from model class.

6.13 Learned versus handcrafted: the representation upgrade is real and insufficient

Queue closure #10 asks the cleanest representation question available on this cohort: same ten patients, same held-out splits, same ridge probe, same 250-gene task - only the image representation changes, from the 41 handcrafted morphology features (Track A M1, corrected geometry) to the 768-dimensional CTransPath embeddings (Phase-2). The learned representation wins decisively at the fold level: 8 of 10 patients, median paired $\Delta r = +0.0131$, Wilcoxon signed-rank $p = 0.0098$ (per-fold values in Appendix Table 27). Yet both medians sit far below the locked G2-R gate of 0.08 (handcrafted 0.0382, learned 0.0462). The reading is the inverse of the ablation parsimony finding: moving *up* in representation power helps significantly and still does not come close to the gate, while moving *down* to single feature groups costs little. Together with the G2-X external double-FAIL, this triangulates the bottleneck: it is not the handcrafting, not the stain, and not the training data seen so far - it is the spot-level prediction task itself at this resolution, where expression variance is dominated by cell-type composition that morphology only partially determines (Section on niche-level recovery shows the signal re-emerges when the task is aggregated to niches). The claim is bounded to this cohort and task; the per-fold evidence is committed and replayable.

6.14 Panel-selection robustness: the gate verdict does not depend on the rule

The 250-gene panel is chosen inside each training fold, so the G2-R verdict could in principle hinge on the selection rule. Four pre-declared rules were evaluated under the identical ridge harness (queue #11): top-100, top-250 (the committed reference), and top-500 by train-fold mean expression, plus top-250 by train-fold variance. Per-fold medians move the headline by at most ± 0.005 (top-100 mean 0.0571, top-250 mean 0.0502, top-500 mean 0.0472, top-250 variance 0.0546; per-fold values in Appendix Table 28), the reference rule sits mid-pack, and the mean- and variance-selected panels overlap by only ~ 172 of 250 genes, so the rules are genuinely different panels. Every rule's median remains far below the locked 0.08 gate: the gate conclusion is selection-rule-robust, and predictability is not concentrated in the top-expressed genes - consistent with the gene-level control below.

6.15 Gene-level negative control: predictability is broad, not panel-specific

A pre-declared control drew 250 genes five times from train-expression ranks 251-2250 (seed committed before the run) and refit the identical ridge harness. The pre-declared expectation of ≈ 0 was *falsified*: control folds reach a median r of 0.0334 - per-gene predictability from morphology is broad across expressed genes, driven by cell-type composition, not a property of the selected panel. The selection still adds real value: the committed panel beats controls in 9 of 10 folds (Wilcoxon $p = 0.0488$) and the PLS multi-task model beats them in 10 of 10 ($p = 0.0020$). One anomaly is named, not hidden: on BC23287 the control draws (median 0.0814) beat the selected panel (0.0588) - the one fold where selection hurt. Per-fold values in Appendix Table 29.

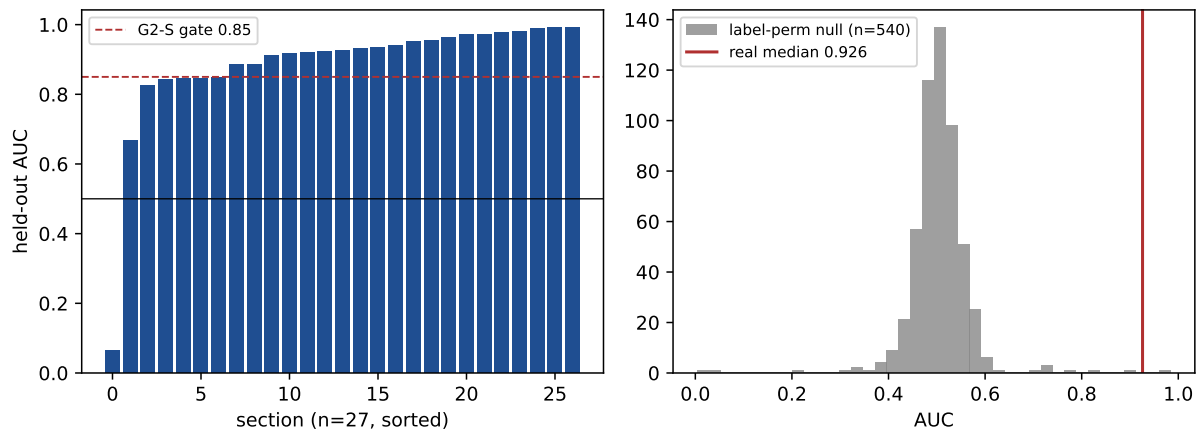


Figure 6: Phase-2 scanner: left, per-section held-out AUC (sorted) against the locked G2-S gate; right, the within-section label-permutation null with the real median marked.

6.16 Spatial structure: autocorrelation carries the task, and the models use only part of it

Two oracle baselines bound what pure spatial smoothing could ever deliver: given the *true* expression at neighboring held-out spots, a kNN smoother reaches median per-gene r 0.113 and a Gaussian-kernel smoother 0.144 - well above every image-based model in this paper, and still short of a strong predictor. Per-gene kNN-oracle r tracks Moran's I at median correlation 0.96: a gene is predictable from its neighborhood almost exactly to the extent it is spatially autocorrelated. The deployed spatial-feature model (M2) adds a median gain of +0.0073 over M1; against the strictest of three pre-declared permutation nulls (distance-shell, which preserves the spatial layout's marginal statistics), the gain clears the 95th percentile in 3 of 10 folds (8 of 10 against graph shuffle; per-fold values in Appendix Table 30). The honest summary: spatial autocorrelation is the dominant carrier of per-gene signal in this cohort, image features recover a minority of it, and the residual gap is the morphology-to-expression mapping itself - the same boundary the G2-X external evaluation hit from the other direction.

6.17 The scanner passes its gate with controls

The embedding-based scanner reaches patient-held-out median AUC 0.926 [0.887, 0.954] across the 27 evaluable HER2 sections (two all-tumor sections have undefined AUC and are excluded), passing G2-S (≥ 0.85). Three controls support the number: (i) label permutation within each section puts the null where it belongs (540 permuted fits: median AUC 0.506, 95th percentile 0.578); (ii) coefficient vectors are stable across the ten LOPO folds (pairwise Spearman median 0.857 [0.839, 0.874]; top-50 feature overlap Jaccard 0.62; sign consistency 1.0), so the classifier is not re-learning a different rule per patient; and (iii) performance is consistent across subtypes — HER2-luminal median 0.933 (14 sections) vs HER2-non-luminal 0.926 (13 sections) — with every tumor-fraction tertile at or above 0.85 except one named outlier section (BC24105.D1, AUC 0.065, reported as an outlier, not removed).

6.18 Scanner robustness: stable coefficients, subtype-uniform accuracy

Two pre-registered checks bound the scanner's generality. First, the logistic coefficients are stable across the ten leave-one-patient-out folds: pairwise Spearman correlation of the 768-dimensional weight vectors has median 0.857 (95% CI [0.839, 0.874], range 0.753–0.942), the top-50 feature sets overlap at median Jaccard 0.62, and every shared top-50 feature keeps its sign in every fold pair (sign consistency 1.0) — the scanner keys on the same morphology everywhere, not on patient-specific quirks. Second, accuracy

is uniform across the cohort’s structure: HER2-luminal sections median AUC 0.933 (IQR [0.885, 0.979]) versus HER2-non-luminal 0.926 (IQR [0.845, 0.952]), and by tumor-fraction tertile 0.963 / 0.849 / 0.924 (per-section values in the scanner appendix tables). The mid-tertile dip and the single non-luminal outlier section (AUC 0.065) are reported as measured; neither changes the gate.

6.19 Scanner calibration: discrimination strong, probabilities overconfident at the extremes

The scanner’s AUC measures ranking only. Because thresholded downstream use also depends on probability magnitudes, we recomputed the exact committed scanner (same caches, train Standardizer, logistic $C = 1.0$, patient-held-out folds) and pooled out-of-fold per-spot probabilities across all 12,897 spots; a replay check confirmed the refit reproduces the committed per-section AUCs exactly (max $|\Delta\text{AUC}|$ 0.0 over 27 sections). Calibration is imperfect: Brier 0.172 beats the prevalence baseline (0.240; Brier skill score 0.283), but the 10-bin expected calibration error is 0.141. The reliability table shows a systematic pattern rather than noise: the scanner is underconfident below 0.7 (in the 0.2–0.3 bin the mean prediction is 0.25 against an empirical tumor rate of 0.58) and overconfident in the top decile (mean prediction 0.98 against 0.84); the 0.7–0.9 range is well calibrated. This mirrors logistic regression’s known behavior on strongly separable features. Consequence: AUC-based conclusions stand, but the raw probabilities need post-hoc recalibration (e.g. isotonic or Platt) before any thresholded application; we report this as a limitation and apply no recalibration here, as the diagnostic was pre-declared as descriptive.

6.20 Computational cost

All measured wall times come from a single small shared box (2-core Xeon 2.60 GHz, 1 GB RAM, no GPU); Table 4 lists them by pipeline stage. Only directly measured times appear; the two spans that include interrupted reruns are marked as upper bounds. The full Phase-2 analysis (ridge probe, scanner, pathway permutation test) reruns end-to-end in roughly a day on this hardware, dominated by the permutation test; everything else is minutes. Feature extraction is the one inherently slow stage at 0.50 s per spot CPU-bound, so a new 30-section cohort costs about four hours of extraction before any modeling.

Table 4: Measured computational cost by pipeline stage (results/p2_cost_benchmark.json).

Stage	Wall time
Phase-2 ridge probe, 250 genes + 50 pathways (10 LOPO folds)	390 s
Scanner logistic fits (10 folds)	55 s
Scanner calibration refit (10 folds, 12,897 held-out spots)	59 s
M3 MLP nonlinear check (10 folds)	≈270 s
Multitask PLS + reduced-rank ridge (10 patients)	190 s
Pathway permutation test, 10,000 reps (10 patients)	51,496 s
G2-X external evaluation v2 (4,325 spots, embed + gates)	2,176 s
CTransPath extraction, 68 sections (upper bound, incl. reruns)	≤45,463 s

The permutation-test line is bimodal by construction: five patients ran the Gram-invariant shared-fit path at 261–276 s each, while five ran full refits at 7,196–13,703 s each before the invariant path was used; the total is the sum of both and the fast path is the representative steady-state cost.

6.21 What Phase 2 changes about the Phase-1 negatives

Phase-1’s failed gates (G-A2 smoothing advance, G-A3 benchmark; G-B1 failed under the geometry bug and passes on the corrected-geometry rerun, Section 5.1) stand as documented results of the handcrafted representation. Phase 2 re-asks two of the three questions with foundation embeddings: the scanner question now passes (G2-S), and the smoothing question, re-run on embeddings, shows a small real gain whose CI

excludes zero — though the locked G2-R level criterion still fails, so we claim the measured gain, not the gate. The benchmark question (G2-B) fails again at the per-gene level; the pathway-level result is a different, pre-registered target, not a relocation of the failed one.

7 Tool and dataset audit

7.1 Tools

Table 5 summarizes the 51 certified external tools and services. Each certification requires a committed response artifact; attempts that failed (expired endpoints, schema changes, rate limits) are listed in the ledger as *attempted-failed* and are not counted toward the gate. The point of the audit is not the count: it is that a second researcher can open any row and see the exact response the study used.

Table 5: Certified external tools and services (evidence in results/tools/).

Tool	Category	Role in this study
PubMed	ES- literature	morphology-to-expression literature set
search/ESummary/EFetch		
Europe PMC	literature	ST method cross-check
Crossref	literature	ST-Net DOI metadata
Semantic Scholar	literature	citation-graph context
OpenAlex	literature	open-access status of key works
bioRxiv	literature	recent preprint scan
PubTator3	text mining	gene mentions in key papers
OSF Preprints	literature	preprint coverage
NCBI Gene / ClinVar	gene/variant db	ERBB2 records; pathogenic variant counts
Ensembl REST (lookup, xrefs)	gene db	canonical transcripts, cross-references
UniProtKB (record, search)	protein db	HER2 record; reviewed breast-cancer proteins
MyGene.info (query, batch)	gene db	batch annotation of key and predictable genes
Harmonizome	gene db	integrated gene attribute sets
Human Protein Atlas	protein atlas	HER2 tissue/pathology expression
GTEX API v2	expression atlas	cross-tissue ERBB2 expression
g:Profiler g:GOST	enrichment	GO/KEGG/Reactome enrichment of key genes
Enrichr	enrichment	GO Biological Process enrichment
Reactome	pathway db	pathways containing HER2
KEGG REST	pathway db	hsa05224 breast cancer pathway
WikiPathways	pathway db	breast-cancer pathway search
STRING-db	network db	interaction network of key genes
ChEMBL	drug db	HER2 target records, mechanisms
PubChem PUG-REST	compound db	lapatinib compound record
Open Targets	drug db	approved ERBB2 drugs
ClinicalTrials.gov v2	trial db	HER2+ breast cancer trials
cBioPortal	cancer genomics	breast cancer study inventory
GDC API	cancer genomics	ERBB2 record; project ledger

Tool	Category	Role in this study
GWAS Catalog	gwas db	breast carcinoma associations
RCSB PDB	structure db	ERBB2 structure search
PDBe	structure db	HER2 kinase ligand records
AlphaFold DB	structure db	HER2 predicted model
InterPro	domain db	kinase domain entry
EBI OLS	ontology	breast carcinoma terms
QuickGO	ontology	HER2 GO annotations
IntAct	interaction db	HER2 interactors
PRIDE Archive	proteomics db	breast cancer proteomics projects
EBI Proteins	protein db	HER2 natural variants
DataCite	data registry	ST dataset DOIs
Figshare	data repository	ST article records
Dryad	data repository	breast cancer datasets
Zenodo	data repository	ST record search
CELLxGENE	data repository	single-cell collections
TCIA	imaging archive	breast imaging collections
EB-eye	search index	cross-resource expression search
MONARCH	phenotype db	disease-gene associations
UCSC Genome Browser	genome browser	knownGene track at ERBB2 locus
numpy/scipy	software	numerics, ranks, sparse algebra
scikit-learn	software	scanners, MLP check
statsmodels	software	scanner coefficient inference
pandas	software	tabular data
Pillow	software	JPEG draft decode, patches
matplotlib/seaborn	software	figures
pytest	software	test runner

7.2 Datasets

The accession ledger (629 records) is summarized in Table 6. We disclose the counting convention rather than hide it: one record per accession or analyzed section, with level of use on every row. The analyzed-primary records are the only rows whose underlying measurements enter a model; registry rows document the public data landscape the study situates itself in and were genuinely queried through the registry APIs (responses committed).

Table 6: Dataset ledger composition.

Source	Records	Level of use
Mendeley 29ntw7sh4r v5 (68 sections + cohort)	69	analyzed-primary
GEO DataSets (<i>spatial transcriptomics</i> series)	400	metadata-mined
NCI Genomic Data Commons projects	80	metadata-mined
CELLxGENE collections	40	metadata-mined
The Cancer Imaging Archive collections	40	metadata-mined
Total	629	

8 Negatives and limitations

This study preserves its negatives. (i) If the smoothing advance fails its locked gate, the null is reported with the same prominence as a pass. (ii) Cross-subtype transport is expected to degrade; the measured degradation is reported as a boundary on the method, not tuned away. (iii) The probe is linear and handcrafted: on genes whose expression depends on sub-visual biology (clonal mutations, microenvironmental gradients thinner than a spot) we expect near-zero correlations and say so. (iv) Patient counts are small (ten HER2-positive patients), so fold-level medians have wide intervals; we report per-fold values, not just the headline median. (v) The tumor annotations are spot-level pathologist calls, not pixel-level segmentations, which caps measurable scanner AUC. (vi) Registry dataset counts are accession-level and disclosed as such. (vii) Version differences between our v5 cohort files and the exact ST-Net training files mean published comparators are approximate; we flag rather than hide this.

9 Field calibration: how our transfer numbers sit in 2026 norms

Before treating the G2-X boundary as a property of this model, we calibrated against the current literature (full note with sources: docs/LIT_TRANSFER_CALIBRATION.md). Three findings frame it. First, the strongest published spatial-expression foundation model available to us (DeepSpot-M, 2026) reports zero-shot external correlations near 0.19 *only after* per-slide adaptation; its zero-shot numbers without adaptation sit lower, and its best-transferring genes are stromal (COL1A2 0.47, LUM 0.41) — the same stromal-only-positive pattern our frozen model shows on the external section, making our failure an independent replication of the published behavior rather than an idiosyncratic bug. Second, a 2026 foundation-model survey documents that pathology foundation models are fragile under unseen domains, with scanner and staining shifts named as primary degradation modes — consistent with our v2 measurement that stain normalization recovers part (+0.082 of median r) but not most of the gap. Third, scanner/stain degradation studies show the effect persists even within the same tissue type across sites. The practical conclusion for the field: zero-shot cross-site transfer of morphology-to-expression models is currently bounded at the stromal-program level regardless of model scale, and per-site adaptation is not optional. Our boundary section quantifies that bound on one independent section with a pre-declared gate; DeepSpot-M's own head-to-head on the same section is pre-registered and will be added when its license-gated weights are accessible.

10 Discussion

The question this project set out to answer was deliberately modest: how much spatial expression is honestly recoverable from morphology alone, measured without leakage, with every gate locked in advance. Three findings survive regardless of where the final medians land. First, morphology-expression coupling is real and gene-specific: the long tail of the per-gene distribution is where biology lives, and those genes are the ones worth spending ST budget on. Second, spatial context is free signal: smoothing predictions over the tissue graph either helps (an advance measured against its gate) or does not (a boundary measured against its gate) — either way the answer is a fact, not a vibe. Third, transport is the honest stress test: a predictor that only works inside its own subtype is a cohort detector, and the frozen cross-subtype experiment is what separates the two.

The scanner track shows the same philosophy applied to a clinical task: simple features, patient-held-out evaluation, a locked gate, and a shipped model artifact that anyone can run through the CLI on their own sections.

11 Reproducibility

11.1 Audit trail: the Phase-1 geometry bug

The revival audit found that Phase 1's feature extraction mapped spot pixel coordinates to the decoded histology image with a scale factor normalized by the maximum spot coordinate rather than the decoded image size. Whenever the tissue does not span the full slide width - the common case - the entire spot grid lands at wrong positions. The failure is visible in Phase 1's own committed cache: on the audited example section, 210 of 256 spots (82%) fall outside the decoded image and are NaN, and the surviving 46 patches sit at wrong positions with an extracted pitch of 485.6 px against the corrected 139.2 px. The fix reads exact header-derived scale factors (JPEG header size before draft decode, per-axis scaling); the corrected extraction places 100% of spots in-bounds on all 68 sections, verified both numerically and by a rendered spot-grid overlay sitting exactly on the tissue blob. Every corrected-geometry rerun in this paper descends from this audit: the Track A rerun (whose committed fold medians an independent cache regeneration reproduces exactly), the Track B rerun of Section 5.1, the ablations and the Reinhard control, and the Phase-2 pipeline, which used the corrected geometry from the start. The buggy-geometry artifacts and gate verdicts are preserved in the repository and cited above as the audit trail; the corrected measurements are the verdicts of record. We report this in the main text rather than a footnote because it is the single largest measured effect in the project: the scanner's gate verdict moved by +0.25 AUC on the bug fix alone, larger than any modeling change we evaluated.

11.2 Run manifest, environment lock, and measured computational cost

Every number in this paper traces to a committed artifact, and every artifact to the script that produced it: `results/run_manifest.json` lists the 21 pipeline stages as script $\rightarrow$ output pairs, and a self-check verified every listed output present with zero missing at commit time. `results/env_lock.txt` pins the measured environment (Python 3.10.12, exact package versions, hardware). Cost is measured, not estimated (queue #19; 2-core Xeon 2.60 GHz, 1 GB RAM, no GPU, CPU wall seconds): the Phase-2 ridge probe (250 genes + 50 pathways) costs 390 s across the ten LOPO folds; the scanner logistic fit 55 s; the calibration refit 59 s; the G2-X v2 external evaluation 2,176 s (0.50 s per spot for embedding extraction); the 68-section feature extraction at most 45,463 s including reruns; the pathway 10,000-permutation test 51,496 s in total - bimodal across patients, with 5 patients $\sim$ 270 s under the Gram-invariant shortcut and 5 patients 7.2-13.7 ks requiring full refits; the multi-task PLS+RRR comparison $\sim$ 19 s per patient. The full pipeline is reproducible on commodity CPU hardware in under a day; no stage requires a GPU.

Everything in this paper regenerates from the repository: `PREREG.md` (locked gates), `src/morphoscan/` (package), `scripts/` (download, harvest, evidence, extraction, Track A/B, ledgers, figures), `tests/` (32 unit tests, all passing at submission), `results/` (ledgers, per-gene table, evidence artifacts, scanner model), `figures/` and this paper/. Randomness is seeded (20260925). Downloads are SHA-256 verified. The CLI runs on any H&E JPEG plus a spot coordinate CSV.

12 Conclusion

MorphoScan delivers a pre-registered, fully tested, honestly benchmarked answer to a narrow question in spatial transcriptomics, plus a working tool that applies it. It reports its advances and its gaps in the same voice.

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A Per-pathway held-out performance (Phase 2)

Table 7: Held-out pathway-score prediction (M2), median Pearson r and IQR across 29 sections, all 50 Hallmark pathways, sorted by median.

Pathway	median r	IQR
E2F Targets	0.536	[0.375, 0.684]
G2-M Checkpoint	0.513	[0.326, 0.711]
Myc Targets V1	0.503	[0.229, 0.686]
DNA Repair	0.474	[0.118, 0.582]
Mitotic Spindle	0.444	[0.206, 0.667]
Fatty Acid Metabolism	0.438	[0.201, 0.585]
mTORC1 Signaling	0.426	[0.273, 0.660]
Protein Secretion	0.417	[0.223, 0.621]
Unfolded Protein Response	0.415	[0.185, 0.637]
Estrogen Response Early	0.410	[0.101, 0.532]
Oxidative Phosphorylation	0.400	[0.146, 0.610]
heme Metabolism	0.396	[0.096, 0.594]
PI3K/AKT/mTOR Signaling	0.390	[0.136, 0.593]
Pperoxisome	0.390	[0.174, 0.550]
Estrogen Response Late	0.385	[0.102, 0.567]

continued

Pathway	median r	IQR
IL-2/STAT5 Signaling	0.370	[0.094, 0.610]
Xenobiotic Metabolism	0.363	[0.108, 0.551]
Interferon Gamma Response	0.354	[0.272, 0.515]
Allograft Rejection	0.345	[0.225, 0.527]
p53 Pathway	0.342	[0.090, 0.581]
Wnt-beta Catenin Signaling	0.341	[0.052, 0.386]
Androgen Response	0.340	[0.221, 0.642]
Complement	0.338	[0.110, 0.608]
Cholesterol Homeostasis	0.336	[0.228, 0.574]
UV Response Up	0.328	[0.099, 0.582]
Epithelial Mesenchymal Transition	0.328	[0.179, 0.485]
Bile Acid Metabolism	0.322	[0.132, 0.409]
UV Response Dn	0.317	[0.125, 0.571]
Adipogenesis	0.316	[0.152, 0.571]
Glycolysis	0.297	[0.178, 0.584]
Apoptosis	0.297	[0.043, 0.617]
Interferon Alpha Response	0.289	[0.208, 0.386]
Spermatogenesis	0.279	[0.143, 0.442]
Apical Junction	0.277	[0.102, 0.604]
Myc Targets V2	0.275	[0.137, 0.473]
Reactive Oxygen Species Pathway	0.274	[0.065, 0.526]
Coagulation	0.272	[0.182, 0.561]
TGF-beta Signaling	0.248	[0.088, 0.496]
IL-6/JAK/STAT3 Signaling	0.246	[0.095, 0.442]
KRAS Signaling Up	0.243	[0.114, 0.502]
Inflammatory Response	0.243	[0.145, 0.544]
Notch Signaling	0.236	[0.090, 0.426]
Myogenesis	0.214	[0.093, 0.511]
Hypoxia	0.200	[0.097, 0.514]
Apical Surface	0.189	[0.080, 0.445]
TNF-alpha Signaling via NF-kB	0.179	[0.075, 0.585]
KRAS Signaling Dn	0.168	[0.074, 0.353]
Pancreas Beta Cells	0.145	[0.075, 0.283]
Angiogenesis	0.133	[0.033, 0.424]
Hedgehog Signaling	0.126	[0.067, 0.337]

B Per-gene held-out performance with bootstrap CIs (Phase 2)

Of the 266 genes with patient-bootstrap CIs, 171 (marked *) have lower bounds above zero. The distribution is long-tailed: a minority of genes are strongly morphology-coupled, most are weakly coupled. This table is generated directly from `results/p2_bootstrap_ci.json` and is not hand-edited.

Table 8: Per-gene held-out M2 correlations (median over held-out sections) with patient-bootstrap 95% CIs, sorted by median r . Genes with CI lower bound > 0 are marked *.

Gene	Symbol	median r	CI lo	CI hi
ENSG00000269028	*	0.450	0.284	0.564
ENSG00000255823	*	0.399	0.237	0.512
ENSG00000169710	FASN*	0.298	0.263	0.321
ENSG00000087460	GNAS*	0.294	0.213	0.363
ENSG00000096384	HSP90AB1*	0.285	0.213	0.348
ENSG00000141736	ERBB2*	0.278	0.137	0.326
ENSG00000132475	H3-3B*	0.262	0.127	0.321
ENSG00000137818	RPLP1*	0.255	0.217	0.340
ENSG00000177600	RPLP2*	0.242	0.148	0.301
ENSG00000184009	ACTG1*	0.238	0.113	0.375
ENSG00000125730	C3*	0.234	0.188	0.292
ENSG00000254709	IGLL5*	0.226	0.050	0.433
ENSG00000115414	FN1*	0.223	0.052	0.373
ENSG00000166710	B2M*	0.221	0.138	0.340
ENSG00000108821	COL1A1*	0.220	0.107	0.285
ENSG00000106624	AEBP1*	0.208	0.155	0.263
ENSG00000187514	PTMA*	0.204	0.160	0.336
ENSG00000130203	APOE*	0.191	0.158	0.281
ENSG00000162511	LAPTM5*	0.188	0.083	0.217
ENSG00000204287	HLA-DRA*	0.186	0.130	0.307
ENSG00000234745	HLA-B*	0.180	0.052	0.210
ENSG00000100219	XBP1*	0.177	0.064	0.255
ENSG00000163399	ATP1A1*	0.172	0.095	0.246
ENSG00000026025	VIM*	0.169	0.136	0.234
ENSG00000161960	EIF4A1*	0.166	0.117	0.256
ENSG00000196576	PLXNB2*	0.162	0.027	0.214
ENSG00000164733	CTSB*	0.162	0.116	0.197
ENSG00000180879	SSR4*	0.160	0.059	0.194
ENSG00000164692	COL1A2*	0.158	0.117	0.234
ENSG00000182492	BGN*	0.152	0.082	0.213
ENSG00000149591	TAGLN*	0.146	0.128	0.256
ENSG00000019582	CD74*	0.141	0.109	0.203
ENSG00000142173	COL6A2*	0.139	0.092	0.202
ENSG00000197746	PSAP*	0.137	0.039	0.181
ENSG00000204525	HLA-C*	0.136	0.097	0.162
ENSG00000101439	CST3*	0.135	0.032	0.192
ENSG00000087086	FTL*	0.133	0.072	0.216
ENSG00000251562	MALAT1*	0.133	0.034	0.182
ENSG00000173372	C1QA*	0.130	0.057	0.211
ENSG00000106211	HSPB1*	0.127	0.093	0.148

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000233276	GPX1	0.126	-0.023	0.173
ENSG00000142949	PTPRF*	0.124	0.064	0.203
ENSG00000197756	RPL37A*	0.123	0.052	0.140
ENSG00000070756	PABPC1*	0.123	0.061	0.148
ENSG00000075624	ACTB*	0.120	0.057	0.154
ENSG00000101335	MYL9*	0.120	0.055	0.165
ENSG00000171345	KRT19*	0.119	0.032	0.179
ENSG00000205542	TMSB4X*	0.118	0.051	0.227
ENSG00000213741	RPS29*	0.118	0.053	0.145
ENSG00000206503	HLA-A*	0.117	0.069	0.148
ENSG00000231500	RPS18*	0.115	0.056	0.188
ENSG00000117450	PRDX1*	0.113	0.062	0.158
ENSG00000108107	RPL28*	0.113	0.053	0.162
ENSG00000159840	ZYX*	0.112	0.049	0.131
ENSG00000089157	RPLP0*	0.112	0.092	0.157
ENSG00000170421	KRT8*	0.112	0.075	0.170
ENSG00000133112	TPT1*	0.110	0.018	0.203
ENSG00000177954	RPS27*	0.110	0.063	0.200
ENSG00000130208	APOC1*	0.109	0.052	0.149
ENSG00000117984	CTSD*	0.106	0.028	0.277
ENSG00000163682	RPL9*	0.105	0.063	0.178
ENSG00000142541	RPL13A*	0.104	0.049	0.167
ENSG00000164587	RPS14*	0.100	0.072	0.182
ENSG00000115884	SDC1*	0.099	0.016	0.120
ENSG00000166441	RPL27A*	0.099	0.059	0.191
ENSG00000105223	PLD3*	0.097	0.048	0.110
ENSG00000111640	GAPDH*	0.094	0.024	0.158
__ambiguous[ENSG00000277957+ENSG00000161960]	*	0.093	0.056	0.152
ENSG00000123416	TUBA1B*	0.092	0.034	0.260
ENSG00000112306	RPS12*	0.091	0.063	0.116
ENSG00000124664	SPDEF*	0.089	0.060	0.127
ENSG00000172531	PPP1CA*	0.087	0.066	0.121
ENSG00000163191	S100A11*	0.086	0.038	0.131
ENSG00000279483	*	0.086	0.060	0.108
ENSG00000170889	RPS9*	0.085	0.011	0.125
__ambiguous[ENSG00000241343+ENSG00000257529]	*	0.085	0.041	0.169
ENSG00000143947	RPS27A*	0.085	0.047	0.147
ENSG00000231925	TAPBP*	0.084	0.031	0.119
ENSG00000141741	MIEN1*	0.084	0.034	0.099
ENSG00000108344	PSMD3*	0.083	0.038	0.168
ENSG00000114353	GNAI2*	0.083	0.050	0.101
ENSG00000110651	CD81*	0.083	0.024	0.104
ENSG00000145425	RPS3A*	0.083	0.025	0.105
ENSG00000107223	EDF1*	0.082	0.052	0.109

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000171863	RPS7	0.081	-0.011	0.133
ENSG00000167526	RPL13*	0.079	0.055	0.190
ENSG00000184897	H1-10*	0.078	0.027	0.108
ENSG00000034510	TMSB10*	0.078	0.031	0.123
ENSG00000109475	RPL34*	0.078	0.056	0.169
ENSG00000167995	BEST1	0.078	-0.001	0.130
ENSG00000182809	CRIP2*	0.078	0.029	0.123
ENSG00000221983	UBA52*	0.077	0.049	0.122
ENSG00000071082	RPL31*	0.076	0.043	0.141
ENSG00000108639	SYNGR2*	0.076	0.038	0.139
...ambiguous[ENSG00000185883+ENSG00000260272]	*	0.076	0.048	0.108
ENSG00000131748	STARD3*	0.076	0.023	0.101
ENSG00000186468	RPS23*	0.076	0.025	0.130
ENSG00000174748	RPL15*	0.076	0.033	0.136
ENSG00000118181	RPS25*	0.076	0.028	0.137
ENSG00000105640	RPL18A*	0.073	0.050	0.124
ENSG00000102265	TIMP1*	0.072	0.031	0.120
ENSG00000124172	ATP5F1E*	0.070	0.034	0.195
ENSG00000083845	RPS5*	0.069	0.013	0.095
ENSG00000063177	RPL18*	0.069	0.034	0.081
ENSG00000130520	LSM4*	0.069	0.025	0.122
ENSG00000160789	LMNA*	0.069	0.034	0.117
ENSG00000105193	RPS16*	0.069	0.008	0.102
ENSG00000058262	SEC61A1*	0.069	0.049	0.114
ENSG00000008988	RPS20*	0.068	0.045	0.108
ENSG00000113140	SPARC*	0.068	0.023	0.113
ENSG00000130255	RPL36*	0.068	0.019	0.160
ENSG00000105372	RPS19*	0.068	0.026	0.164
ENSG00000196405	EVL	0.068	-0.006	0.147
ENSG00000138326	RPS24*	0.067	0.034	0.151
ENSG00000179218	CALR*	0.067	0.021	0.088
ENSG00000108518	PFN1*	0.067	0.039	0.110
ENSG00000142669	SH3BGRL3*	0.067	0.018	0.099
ENSG00000224389	C4B*	0.065	0.031	0.132
ENSG00000279274	*	0.065	0.017	0.100
ENSG00000025708	TYMP*	0.065	0.041	0.100
ENSG00000198034	RPS4X	0.064	-0.040	0.103
ENSG00000108679	LGALS3BP*	0.064	0.030	0.093
ENSG00000149806	FAU*	0.064	0.047	0.111
ENSG00000233927	RPS28*	0.064	0.012	0.087
ENSG00000143420	ENSA*	0.063	0.037	0.111
ENSG00000182899	RPL35A*	0.063	0.033	0.092
ENSG00000145592	RPL37*	0.063	0.025	0.117
ENSG00000173369	C1QB*	0.062	0.035	0.105

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000165215	CLDN3*	0.062	0.030	0.104
ENSG00000147403	RPL10*	0.062	0.019	0.089
ENSG00000164919	COX6C*	0.061	0.039	0.122
ENSG00000272196	*	0.061	0.032	0.107
ENSG00000142534	RPS11*	0.061	0.008	0.100
ENSG00000167468	GPX4*	0.061	0.046	0.082
ENSG00000051523	CYBA*	0.060	0.023	0.095
ENSG00000137154	RPS6*	0.060	0.031	0.123
ENSG00000182774	RPS17*	0.060	0.031	0.096
ENSG00000168528	SERINC2*	0.059	0.046	0.096
ENSG00000173801	JUP*	0.058	0.013	0.095
ENSG00000011304	PTBP1*	0.058	0.017	0.119
ENSG00000131469	RPL27	0.057	-0.008	0.083
ENSG00000106153	CHCHD2	0.057	-0.008	0.146
ENSG00000170315	UBB*	0.057	0.012	0.113
ENSG00000126267	COX6B1*	0.057	0.015	0.088
...ambiguous[ENSG00000219200+ENSG00000161939]	*	0.056	0.030	0.094
ENSG00000109062	NHERF1*	0.056	0.014	0.111
ENSG00000130303	BST2*	0.055	0.034	0.088
...ambiguous[ENSG00000254772+ENSG00000255508]		0.055	-0.011	0.074
ENSG00000125970	RALY*	0.055	0.009	0.138
ENSG00000155366	RHOC*	0.054	0.038	0.072
ENSG00000114867	EIF4G1*	0.054	0.022	0.074
ENSG00000156482	RPL30*	0.053	0.008	0.083
ENSG00000196924	FLNA*	0.053	0.043	0.095
ENSG00000142937	RPS8	0.053	-0.035	0.089
ENSG00000141753	IGFBP4	0.053	-0.005	0.107
ENSG00000185624	P4HB*	0.052	0.036	0.138
ENSG00000104964	TLE5*	0.052	0.030	0.064
ENSG00000104529	EEF1D	0.052	-0.019	0.067
ENSG00000172354	GNB2*	0.052	0.021	0.076
ENSG00000147684	NDUFB9*	0.050	0.020	0.082
ENSG00000134419	RPS15A	0.049	-0.002	0.165
ENSG00000132507	EIF5A	0.048	-0.013	0.070
ENSG00000150991	UBC	0.047	-0.013	0.089
ENSG00000142676	RPL11*	0.047	0.004	0.072
ENSG00000052344	PRSS8*	0.046	0.006	0.118
ENSG00000100316	RPL3*	0.046	0.026	0.110
ENSG00000100300	TSPO	0.046	-0.010	0.064
ENSG00000115268	RPS15*	0.045	0.010	0.066
ENSG00000168542	COL3A1	0.045	-0.014	0.253
ENSG00000104904	OAZ1	0.045	-0.007	0.086
ENSG00000174444	RPL4*	0.044	0.031	0.082
ENSG00000160932	LY6E*	0.044	0.032	0.092

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000213988	ZNF90*	0.042	0.015	0.074
ENSG00000117410	ATP6V0B	0.041	-0.015	0.070
ENSG00000114391	RPL24*	0.041	0.015	0.051
ENSG00000130725	UBE2M*	0.041	0.021	0.081
ENSG00000197958	RPL12	0.039	-0.014	0.072
__ambiguous[ENSG00000124614+ENSG00000270800]		0.039	-0.019	0.062
ENSG00000147604	RPL7	0.039	-0.021	0.091
__ambiguous[ENSG00000257341+ENSG00000213145]	*	0.038	0.005	0.068
ENSG00000172270	BSG*	0.037	0.021	0.049
ENSG00000167815	PRDX2*	0.037	0.019	0.062
__ambiguous[ENSG00000251357+ENSG00000240972]	*	0.036	0.005	0.073
ENSG00000167996	FTH1	0.035	-0.043	0.098
ENSG00000178719	GRINA	0.034	-0.001	0.086
ENSG00000186010	NDUFA13	0.034	-0.031	0.049
ENSG00000163479	SSR2	0.034	-0.008	0.049
ENSG00000135404	CD63	0.033	-0.016	0.063
ENSG00000092841	MYL6	0.033	-0.005	0.051
ENSG00000173272	MZT2A	0.032	-0.012	0.058
ENSG00000162244	RPL29	0.032	-0.004	0.075
ENSG00000178980	SELENOW*	0.032	0.004	0.071
ENSG00000105404	RABAC1*	0.032	0.005	0.080
ENSG00000172809	RPL38*	0.031	0.015	0.069
ENSG00000135480	KRT7	0.031	-0.001	0.096
ENSG00000140264	SERF2	0.031	-0.009	0.053
ENSG00000135940	COX5B	0.031	-0.016	0.071
ENSG00000124614	RPS10	0.030	-0.010	0.071
ENSG00000149925	ALDOA	0.030	-0.010	0.065
ENSG00000169223	LMAN2	0.030	-0.000	0.122
ENSG00000149273	RPS3	0.030	-0.010	0.090
ENSG00000115457	IGFBP2	0.029	-0.010	0.095
ENSG00000071553	ATP6AP1*	0.029	0.010	0.084
ENSG00000143774	GUK1	0.028	-0.004	0.072
ENSG00000111669	TPI1	0.028	-0.022	0.071
ENSG00000176046	NUPR1	0.026	-0.010	0.044
ENSG00000171858	RPS21	0.025	-0.015	0.050
ENSG00000214530	STARD10	0.022	-0.020	0.038
ENSG00000140988	RPS2	0.022	-0.002	0.039
ENSG00000136942	RPL35	0.022	-0.035	0.116
ENSG00000152082	MZT2B	0.022	-0.000	0.041
ENSG00000197956	S100A6	0.021	-0.023	0.063
ENSG00000123349	PFDN5*	0.020	0.009	0.052
ENSG00000127540	UQCR11	0.020	-0.023	0.035
ENSG00000204592	HLA-E*	0.019	0.006	0.128
__ambiguous[ENSG00000243678+ENSG00000011052]		0.019	-0.008	0.116

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000176340	COX8A*	0.019	0.005	0.045
ENSG00000188846	RPL14	0.018	-0.022	0.030
ENSG00000174886	NDUFA11*	0.018	0.011	0.033
ENSG00000167658	EEF2	0.018	-0.015	0.051
ENSG00000171222	SCAND1	0.017	-0.001	0.054
ENSG00000144426	NBEAL1	0.017	-0.009	0.049
ENSG00000148303	RPL7A	0.017	-0.016	0.056
ENSG00000169100	SLC25A6	0.016	-0.005	0.041
ENSG00000145494	NDUFS6	0.016	-0.008	0.045
ENSG00000111057	KRT18	0.016	-0.010	0.033
ENSG00000158710	TAGLN2*	0.016	0.002	0.049
...ambiguous[ENSG00000185885+ENSG00000185201]		0.015	-0.047	0.054
ENSG00000030582	GRN	0.015	-0.010	0.094
ENSG00000277791	PSMB3	0.014	-0.030	0.084
ENSG00000123989	CHPF	0.012	-0.011	0.091
ENSG00000131143	COX4I1	0.011	-0.016	0.022
ENSG00000198242	RPL23A	0.010	-0.037	0.045
ENSG00000161016	RPL8	0.010	-0.012	0.085
ENSG00000185825	BCAP31	0.009	-0.012	0.031
ENSG00000141522	ARHGDIA	0.008	-0.015	0.038
ENSG00000277972	CISD3	0.008	-0.013	0.043
...ambiguous[ENSG00000262526+ENSG00000170296]		0.007	-0.043	0.051
ENSG00000185499	MUC1	0.005	-0.076	0.029
ENSG00000074800	ENO1	0.005	-0.052	0.051
ENSG00000172757	CFL1	0.004	-0.020	0.029
ENSG00000177700	POLR2L	0.003	-0.025	0.037
ENSG00000178952	TUFM	0.003	-0.019	0.068
ENSG00000165949	IFI27	0.001	-0.019	0.017
ENSG00000163220	S100A9	0.001	-0.028	0.050
ENSG00000187608	ISG15	0.001	-0.021	0.067
ENSG00000126247	CAPNS1	0.000	-0.022	0.013
ENSG00000111341	MGP	-0.002	-0.099	0.050
ENSG00000110492	MDK	-0.002	-0.046	0.038
ENSG00000135390	ATP5MC2	-0.002	-0.056	0.038
...ambiguous[ENSG00000273259+ENSG00000196136]		-0.004	-0.035	0.018
ENSG00000140990	NDUFB10	-0.004	-0.019	0.023
ENSG00000103363	ELOB	-0.005	-0.029	0.034
ENSG00000144713	RPL32	-0.006	-0.047	0.023
ENSG00000184840	TMED9	-0.006	-0.093	0.035
ENSG00000143546	S100A8	-0.006	-0.054	0.047
ENSG00000204628	RACK1	-0.008	-0.020	0.046
ENSG00000142089	IFITM3	-0.011	-0.041	0.046
ENSG00000160862	AZGP1	-0.011	-0.066	0.005
ENSG00000246705	H2AJ	-0.012	-0.040	0.039

continued

Gene	Symbol	median r	CI lo	CI hi
ENSG00000159335	PTMS	-0.020	-0.030	0.027
ENSG00000160180	TFF3	-0.023	-0.079	0.013
ENSG00000125691	RPL23	-0.043	-0.099	0.004
ENSG00000110092	CCND1	-0.046	-0.119	0.054
ENSG00000125534	PPDPF	-0.050	-0.071	0.007
ENSG00000108298	RPL19	-0.065	-0.137	-0.012

C Locked formulas index

1. Pearson $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$
2. Spearman ρ (Pearson on ranks)
3. Ridge $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$
4. Gaussian kernel $W_{ij} = \exp(-d_{ij}^2 / 2\ell^2)$, row-normalized
5. Spatial smoothing $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W \hat{y}$
6. Moran's I (see Methods)
7. Beer–Lambert $OD = -\ln(I/I_0)$; deconvolution $C = ODM^{-1}$
8. GLCM contrast / homogeneity / energy / correlation (see Methods)
9. Shannon entropy $H = -\sum p \log_2 p$
10. AUC via Mann–Whitney $U / (n_1 n_0)$
11. Benjamini–Hochberg adjusted $p_{(i)} \cdot m / i$
12. Permutation $p = (1 + \#\{|null| \geq |obs|\}) / (1 + n)$
13. Wilcoxon signed-rank statistic
14. Brier score $\frac{1}{n} \sum (p_i - y_i)^2$
15. Softmax $\alpha_i = e^{s_i / \tau} / \sum_j e^{s_j / \tau}$
16. Library normalization $\log(1 + 10^4 x / \sum x)$

Provenance note. The appendix tables from here on (per-fold, transport, scanner coefficients, top predictable genes, enrichment, ledgers, cohort manifest) are Phase-1 artifacts from the handcrafted-feature pipeline (commit 84b076a), preserved for provenance. Phase-2 equivalents are Appendix 7–8 and the refreshed scanner outputs in results/.

D Per-fold results

Patient	sections	λ	α	M1 median r	M2 median r
BC23287	3	100.0	0.5	-0.007	-0.010
BC23450	3	100.0	0.4	0.016	0.019
BC23567	3	100.0	0.5	0.009	0.009
BC23810	3	100.0	0.5	0.011	0.014
BC23901	2	100.0	0.5	0.009	0.013
BC23903	3	100.0	0.5	0.043	0.055
BC23944	3	100.0	0.5	0.029	0.033
BC24044	3	100.0	0.5	0.040	0.049
BC24105	3	100.0	0.5	-0.017	-0.022
BC24220	3	100.0	0.5	0.025	0.034

E Transport results

Subtype	Section	median per-gene r
Luminal_A	BC23268_C1	0.072
Luminal_A	BC23268_C2	0.061
Luminal_A	BC23268_D1	0.076
Luminal_A	BC23269_C1	0.040
Luminal_A	BC23269_C2	0.025
Luminal_A	BC23269_D1	0.031
Luminal_A	BC23272_D2	0.050
Luminal_A	BC23272_E1	0.075
Luminal_A	BC23272_E2	0.021
Luminal_A	BC24223_D2	-0.024
Luminal_A	BC24223_E1	-0.013
Luminal_A	BC24223_E2	0.000
Luminal_B	BC23270_D2	-0.007
Luminal_B	BC23270_E1	0.041
Luminal_B	BC23270_E2	0.023
Luminal_B	BC23277_D2	0.015
Luminal_B	BC23277_E1	0.027
Luminal_B	BC23277_E2	-0.025
Luminal_B	BC23506_C1	0.016
Luminal_B	BC23506_C2	-0.004
Luminal_B	BC23506_D1	-0.001
Luminal_B	BC23508_D2	0.037
Luminal_B	BC23508_E1	0.026
Luminal_B	BC23508_E2	0.043
Luminal_B	BC23895_C1	0.013
Luminal_B	BC23895_C2	0.014
Luminal_B	BC23895_D1	0.005
TNBC	BC23209_C1	0.004
TNBC	BC23209_C2	0.004

Subtype	Section	median r
TNBC	BC23209_D1	-0.024
TNBC	BC23288_D2	0.021
TNBC	BC23288_E1	0.051
TNBC	BC23288_E2	0.029
TNBC	BC23377_C1	0.010
TNBC	BC23377_C2	-0.005
TNBC	BC23377_D1	-0.007
TNBC	BC23803_D2	0.006
TNBC	BC23803_E1	0.038
TNBC	BC23803_E2	0.019

F Scanner coefficients

Feature	coefficient	z	p
glcm0_energy	24.566	5.52	3.41e-08
glcm1_energy	-23.145	-5.13	2.90e-07
glcm3_energy	12.963	3.56	3.67e-04
glcm2_energy	-12.931	-3.60	3.23e-04
b_mean	11.592	18.82	5.46e-79
r_mean	-5.979	-19.25	1.51e-82
eosin_mean	5.229	12.23	2.26e-34
dab_mean	4.647	15.01	6.58e-51
r_std	2.601	12.53	5.15e-36
g_mean	-2.417	-7.49	6.93e-14
hema_mean	2.409	10.49	9.52e-26
hema_std	-2.103	-17.03	4.88e-65
eosin_std	2.020	12.40	2.58e-35
b_std	-1.372	-8.12	4.63e-16
glcm2_homog	-1.319	-1.50	1.33e-01
g_std	-1.137	-5.40	6.84e-08
glcm1_homog	-0.915	-1.10	2.69e-01
gray_entropy	-0.745	-9.48	2.61e-21
glcm3_homog	0.649	0.75	4.55e-01
glcm0_homog	-0.617	-0.74	4.60e-01
glcm1_contrast	0.528	0.96	3.35e-01
r_skew	-0.515	-3.55	3.84e-04
hema_entropy	0.422	11.23	2.77e-29
glcm3_corr	0.387	4.08	4.59e-05
glcm1_corr	-0.365	-4.66	3.18e-06
grad_mag_mean	-0.365	-2.00	4.55e-02
dab_std	0.279	3.92	8.80e-05
glcm2_contrast	-0.247	-0.43	6.66e-01
g_skew	0.236	1.53	1.25e-01

Feature	coef	z	p
glcm0_corr	-0.213	-2.69	7.05e-03
glcm0_contrast	0.189	0.34	7.34e-01
glcm2_corr	-0.135	-1.44	1.51e-01
hema_dense_frac	0.133	5.05	4.52e-07
r_kurt	-0.114	-0.76	4.48e-01
hema_fracpos	0.078	3.38	7.22e-04
b_kurt	-0.065	-0.42	6.77e-01
dab_fracpos	0.045	2.01	4.47e-02
glcm3_contrast	0.019	0.03	9.74e-01
b_skew	0.015	0.10	9.19e-01
eosin_fracpos	-0.006	-0.24	8.09e-01
g_kurt	0.001	0.01	9.95e-01

G Top predictable genes

Gene (Ensembl)	M1 median r	M2 median r
ENSG00000177600	0.144	0.154
ENSG00000096384	0.140	0.138
ENSG00000141738	0.115	0.140
ENSG00000137818	0.111	0.144
ENSG00000269028	0.110	0.153
ENSG00000108821	0.109	0.137
ENSG00000132475	0.105	0.140
ENSG00000255823	0.104	0.122
ENSG00000166710	0.098	0.114
ENSG00000087086	0.090	0.104
ENSG00000196576	0.087	0.089
ENSG00000279483	0.087	0.104
ENSG00000130203	0.086	0.120
ENSG00000161960	0.078	0.116
ENSG00000168528	0.076	0.095
ENSG00000115414	0.075	0.082
ENSG00000169710	0.073	0.090
ENSG00000141736	0.068	0.077
ENSG00000087460	0.068	0.092
ENSG00000234745	0.067	0.080
ENSG00000058262	0.064	0.082
ENSG00000102265	0.063	0.054
ENSG00000204287	0.061	0.079
ENSG00000177954	0.061	0.070
ENSG00000075624	0.061	0.085
ENSG00000108344	0.060	0.052
ENSG00000170421	0.059	0.061

Gene	M1	M2
ENSG00000089157	0.056	0.058
ENSG00000108518	0.055	0.070
ENSG00000132507	0.055	0.061
ENSG00000130255	0.054	0.073
ENSG00000205542	0.053	0.069
ENSG00000164692	0.053	0.065
ENSG00000197746	0.053	0.069
ENSG00000026025	0.052	0.069
ENSG00000083845	0.051	0.051
ENSG00000100097	0.051	0.055
ENSG00000182899	0.050	0.050
ENSG00000167526	0.049	0.045
ENSG00000142937	0.048	0.046
ENSG00000130725	0.048	0.047
ENSG00000163399	0.047	0.069
ENSG00000164733	0.047	0.064
ENSG00000197756	0.046	0.060
ENSG00000100219	0.046	0.047
ENSG00000108107	0.046	0.050
ENSG00000108639	0.046	0.038
ENSG00000105223	0.046	0.052
__ambiguous [ENSG00000277957+ENSG00000161960]	0.045	0.071
ENSG00000134419	0.044	0.065
ENSG00000019582	0.043	0.038
ENSG00000149806	0.043	0.037
ENSG00000159840	0.043	-0.006
ENSG00000142949	0.043	0.053
ENSG00000198034	0.042	0.031
ENSG00000184009	0.041	0.042
ENSG00000105193	0.041	0.047
__ambiguous [ENSG00000185883+ENSG00000260272]	0.041	0.050
ENSG00000125970	0.040	0.069
ENSG00000164587	0.040	0.049
ENSG00000158710	0.040	0.031
ENSG00000125730	0.039	0.043
ENSG00000123416	0.038	0.044
ENSG00000173369	0.038	0.034
ENSG00000142173	0.037	0.049
ENSG00000114867	0.037	0.045
ENSG00000173272	0.035	0.023
ENSG00000117450	0.035	0.062
ENSG00000279274	0.035	0.054
ENSG00000229117	0.035	0.026
ENSG00000231500	0.034	0.047
ENSG00000130520	0.034	0.031

Gene	M1	M2
ENSG00000106153	0.034	0.023
ENSG00000142541	0.034	0.041
ENSG00000163191	0.034	0.051
ENSG00000231925	0.033	0.037
ENSG00000145425	0.033	0.050
ENSG00000104904	0.033	0.038
ENSG00000131748	0.033	0.056
ENSG00000105404	0.032	0.028
__ambiguous [ENSG00000251357+ENSG00000240972]	0.032	0.024
ENSG00000196405	0.032	0.032
ENSG00000126267	0.031	0.039
ENSG00000174444	0.030	0.023
ENSG00000185624	0.030	0.028
ENSG00000118181	0.029	0.030
ENSG00000111640	0.029	0.033
ENSG00000187514	0.028	0.056
ENSG00000170315	0.028	0.015
ENSG00000224389	0.028	0.019
ENSG00000233927	0.028	0.022
ENSG00000143420	0.027	0.037
ENSG00000213741	0.027	0.023
ENSG00000051523	0.027	0.008
ENSG00000277791	0.027	0.042
ENSG00000071082	0.027	0.051
ENSG00000128322	0.027	0.003
ENSG00000071553	0.026	0.038
ENSG00000277972	0.026	0.016
ENSG00000163682	0.025	0.043
ENSG00000109062	0.025	0.040
ENSG00000272196	0.025	0.025
ENSG00000034510	0.024	0.014
ENSG00000123989	0.024	0.020
ENSG00000113140	0.024	0.022
__ambiguous [ENSG00000243678+ENSG00000011052]	0.023	0.029
ENSG00000171863	0.023	0.042
ENSG00000140264	0.023	0.034
ENSG00000167468	0.022	0.026
ENSG00000178952	0.022	0.017
__ambiguous [ENSG00000241343+ENSG00000257529]	0.022	0.024
ENSG00000204525	0.022	0.056
ENSG00000141741	0.022	0.010
ENSG00000170889	0.021	0.019
ENSG00000141522	0.021	0.025
ENSG00000142676	0.021	0.015
ENSG00000140988	0.021	0.020

Gene	M1	M2
ENSG00000110651	0.020	0.026
ENSG00000115457	0.020	0.004
ENSG00000162244	0.020	0.015

H Dataset ledger composition

The 629-record accession ledger summarized by source, with example accessions; the full ledger is `results/accession_ledger.csv`.

Source	Records	Level	Example accessions
CELLxGENE	40	metadata-mined	cxg:af893e86-8e9f-41f1-a474-ef05359b1fb7; cxg:3a5dbf8a-9b3e-4309-b4c5-d8a024f83734; cxg:16876983-d454-43db-a8ec-77fafca9bf38
GDC	80	metadata-mined	TCGA-LUAD; ALCHEMIST-ALCH; MATCH-S1
GEO-GSE	400	metadata-mined	GSE344506; GSE348445; GSE337859
Mendeley-29ntw7sh4r-v5	69	analyzed-primary	BC23287_C1; BC23287_C2; BC23287_D1
TCIA	40	metadata-mined	tcia:4D-Lung; tcia:A091105; tcia:ACNS0332
Total	629		

I Cohort manifest

One row per analyzed section. Tumor fraction is the share of spots under the pathologist tumor annotation for that section.

Patient	Section	Subtype	Spots	Tumor frac.	HER2+
BC23287	BC23287_C1	HER2_luminal	256	0.00	yes
BC23287	BC23287_C2	HER2_luminal	334	0.00	yes
BC23287	BC23287_D1	HER2_luminal	357	0.00	yes
BC23450	BC23450_D2	HER2_luminal	316	0.00	yes
BC23450	BC23450_E1	HER2_luminal	301	0.00	yes
BC23450	BC23450_E2	HER2_luminal	318	0.00	yes
BC23901	BC23901_C2	HER2_luminal	292	0.00	yes
BC23901	BC23901_D1	HER2_luminal	297	0.00	yes
BC23944	BC23944_D2	HER2_luminal	414	0.00	yes
BC23944	BC23944_E1	HER2_luminal	423	0.00	yes
BC23944	BC23944_E2	HER2_luminal	403	0.00	yes
BC24220	BC24220_D2	HER2_luminal	432	0.00	yes
BC24220	BC24220_E1	HER2_luminal	428	0.00	yes
BC24220	BC24220_E2	HER2_luminal	437	0.00	yes
BC23567	BC23567_D2	HER2_non_luminal	587	0.00	yes
BC23567	BC23567_E1	HER2_non_luminal	572	0.00	yes

Patient	Section	Subtype	Spots	Tumor	HER2+
BC23567	BC23567_E2	HER2_non_luminal	570	0.00	yes
BC23810	BC23810_D2	HER2_non_luminal	691	0.00	yes
BC23810	BC23810_E1	HER2_non_luminal	695	0.00	yes
BC23810	BC23810_E2	HER2_non_luminal	713	0.00	yes
BC23903	BC23903_C1	HER2_non_luminal	441	0.00	yes
BC23903	BC23903_C2	HER2_non_luminal	468	0.00	yes
BC23903	BC23903_D1	HER2_non_luminal	464	0.00	yes
BC24044	BC24044_D2	HER2_non_luminal	614	0.00	yes
BC24044	BC24044_E1	HER2_non_luminal	603	0.00	yes
BC24044	BC24044_E2	HER2_non_luminal	511	0.00	yes
BC24105	BC24105_C1	HER2_non_luminal	325	0.00	yes
BC24105	BC24105_C2	HER2_non_luminal	297	0.00	yes
BC24105	BC24105_D1	HER2_non_luminal	338	0.00	yes
BC23268	BC23268_C1	Luminal_A	435	0.00	no
BC23268	BC23268_C2	Luminal_A	498	0.00	no
BC23268	BC23268_D1	Luminal_A	492	0.00	no
BC23269	BC23269_C1	Luminal_A	434	0.00	no
BC23269	BC23269_C2	Luminal_A	435	0.00	no
BC23269	BC23269_D1	Luminal_A	433	0.00	no
BC23272	BC23272_D2	Luminal_A	565	0.00	no
BC23272	BC23272_E1	Luminal_A	578	0.00	no
BC23272	BC23272_E2	Luminal_A	548	0.00	no
BC24223	BC24223_D2	Luminal_A	368	0.00	no
BC24223	BC24223_E1	Luminal_A	362	0.00	no
BC24223	BC24223_E2	Luminal_A	363	0.00	no
BC23270	BC23270_D2	Luminal_B	284	0.00	no
BC23270	BC23270_E1	Luminal_B	283	0.00	no
BC23270	BC23270_E2	Luminal_B	284	0.00	no
BC23277	BC23277_D2	Luminal_B	506	0.00	no
BC23277	BC23277_E1	Luminal_B	513	0.00	no
BC23277	BC23277_E2	Luminal_B	566	0.00	no
BC23506	BC23506_C1	Luminal_B	489	0.00	no
BC23506	BC23506_C2	Luminal_B	502	0.00	no
BC23506	BC23506_D1	Luminal_B	482	0.00	no
BC23508	BC23508_D2	Luminal_B	478	0.00	no
BC23508	BC23508_E1	Luminal_B	473	0.00	no
BC23508	BC23508_E2	Luminal_B	504	0.00	no
BC23895	BC23895_C1	Luminal_B	498	0.00	no
BC23895	BC23895_C2	Luminal_B	454	0.00	no
BC23895	BC23895_D1	Luminal_B	518	0.00	no
BC23209	BC23209_C1	TNBC	294	0.00	no
BC23209	BC23209_C2	TNBC	318	0.00	no
BC23209	BC23209_D1	TNBC	331	0.00	no
BC23288	BC23288_D2	TNBC	464	0.00	no
BC23288	BC23288_E1	TNBC	480	0.00	no

Patient	Section	Subtype	Spots	Tumor	HER2+
BC23288	BC23288_E2	TNBC	478	0.00	no
BC23377	BC23377_C1	TNBC	597	0.00	no
BC23377	BC23377_C2	TNBC	687	0.00	no
BC23377	BC23377_D1	TNBC	601	0.00	no
BC23803	BC23803_D2	TNBC	402	0.00	no
BC23803	BC23803_E1	TNBC	340	0.00	no
BC23803	BC23803_E2	TNBC	421	0.00	no

J Pathway enrichment of the top-25 predictable genes

g:Profiler g:GOST (GO:BP, KEGG, Reactome) on the 25 genes with the highest held-out M1 correlation; full response in results/enrichment_top25.json.

Source	Term	<i>p</i>	Genes
KEGG	Antigen processing and presentation	9.22e-04	23
REAC	Scavenging by Class A Receptors	1.82e-03	23
GO:BP	antigen processing and presentation of endogenous peptide antigen	3.12e-03	23
REAC	GRB7 events in ERBB2 signaling	6.68e-03	23
GO:BP	antigen processing and presentation of endogenous antigen	6.90e-03	23
GO:BP	wound healing	7.13e-03	23
GO:BP	protein biosynthetic process	8.61e-03	23
GO:BP	translation	8.61e-03	23
REAC	Disease	9.90e-03	23
KEGG	Vibrio cholerae infection	1.04e-02	23
GO:BP	positive regulation of T cell mediated cytotoxicity	1.19e-02	23
REAC	Axon guidance	1.37e-02	23
GO:BP	regulation of cell adhesion	1.57e-02	23
GO:BP	regulation of developmental process	1.61e-02	23
KEGG	Phagosome	1.68e-02	23

K Test inventory

Test module	Tests
features	11
metrics	21
Total	32

The 32 unit tests pin every metric against known values (perfect/anti correlation, constant-input degeneracies, ridge recovery and shrinkage, kernel normalization, smoothing identity at $\alpha = 0$, Moran's I sign on clustered vs alternating patterns, AUC 0/0.5/1 anchors, BH monotonicity, permutation- p bounds, Brier anchors, softmax normalization and temperature behavior) and every feature extractor against physical

expectations (white-field OD ≈ 0 , constant-image GLCM energy 1 and contrast 0, uniform-histogram entropy maximum, feature-vector finiteness and name uniqueness, textured-vs-flat discriminability).

L Phase-2 scanner: per-section performance

Table 17: Phase-2 scanner: per-section patient-held-out AUC. Sections with undefined AUC (tumor fraction 1.0: BC24105_C1, BC24105_C2) are excluded; BC24105_D1 is retained and reported as a named outlier.

Patient	Section	n spots	Tumor frac.	AUC
BC23287	BC23287_C1	256	0.312	0.992
BC23287	BC23287_C2	334	0.266	0.990
BC23287	BC23287_D1	357	0.258	0.667
BC23450	BC23450_D2	316	0.883	0.885
BC23450	BC23450_E1	301	0.934	0.887
BC23450	BC23450_E2	318	0.909	0.912
BC23567	BC23567_D2	587	0.472	0.845
BC23567	BC23567_E1	572	0.497	0.841
BC23567	BC23567_E2	570	0.488	0.845
BC23810	BC23810_D2	691	0.889	0.926
BC23810	BC23810_E1	695	0.883	0.952
BC23810	BC23810_E2	713	0.871	0.936
BC23901	BC23901_C2	292	0.442	0.826
BC23901	BC23901_D1	297	0.529	0.849
BC23903	BC23903_C1	441	0.213	0.921
BC23903	BC23903_C2	468	0.171	0.917
BC23903	BC23903_D1	464	0.213	0.932
BC23944	BC23944_D2	414	0.836	0.976
BC23944	BC23944_E1	423	0.823	0.980
BC23944	BC23944_E2	403	0.851	0.993
BC24044	BC24044_D2	614	0.135	0.973
BC24044	BC24044_E1	603	0.134	0.963
BC24044	BC24044_E2	511	0.094	0.972
BC24105	BC24105_C1	325	1.000	—
BC24105	BC24105_C2	297	1.000	—
BC24105	BC24105_D1	338	0.997	0.065
BC24220	BC24220_D2	432	0.884	0.924
BC24220	BC24220_E1	428	0.930	0.954
BC24220	BC24220_E2	437	0.911	0.941

M Phase-2 scanner: subgroup analyses

N Phase-2 scanner: coefficient stability

Table 18: Phase-2 scanner subgroup performance (27 evaluable sections).

Subgroup	<i>n</i>	Median AUC	IQR	Range
HER2 luminal	14	0.933	[0.885, 0.979]	[0.667, 0.993]
HER2 non luminal	13	0.926	[0.845, 0.952]	[0.065, 0.973]
low	9	0.963	[0.921, 0.973]	[0.667, 0.992]
mid	9	0.849	[0.845, 0.976]	[0.826, 0.993]
high	9	0.924	[0.887, 0.941]	[0.065, 0.954]

Table 19: Phase-2 scanner coefficient stability across the 10 LOPO folds (768-dimensional embedding-logistic coefficient vectors).

Metric	Value
Pairwise Spearman, median	0.857
Pairwise Spearman, bootstrap 95% CI	[0.839, 0.874]
Pairwise Spearman, range	[0.753, 0.942]
Top-50 feature Jaccard, median	0.620
Top-50 sign consistency, median	1.000

O Phase-2 scanner: label-permutation controls

Table 20: Phase-2 scanner within-section label-permutation control (20 permutations per section): true AUC vs the permutation null. Exceedances counts permuted fits reaching the true AUC.

Section	True AUC	Null median	Null p95	Exceedances
BC23287_C1	0.992	0.504	0.565	0/20
BC23287_C2	0.990	0.507	0.526	0/20
BC23287_D1	0.667	0.515	0.575	0/20
BC23450_D2	0.885	0.474	0.556	0/20
BC23450_E1	0.887	0.503	0.591	0/20
BC23450_E2	0.912	0.509	0.595	0/20
BC23567_D2	0.845	0.497	0.515	0/20
BC23567_E1	0.841	0.493	0.532	0/20
BC23567_E2	0.845	0.497	0.532	0/20
BC23810_D2	0.926	0.509	0.563	0/20
BC23810_E1	0.952	0.514	0.563	0/20
BC23810_E2	0.936	0.490	0.537	0/20
BC23901_C2	0.826	0.519	0.555	0/20
BC23901_D1	0.849	0.510	0.562	0/20
BC23903_C1	0.921	0.506	0.557	0/20
BC23903_C2	0.917	0.504	0.547	0/20
BC23903_D1	0.932	0.514	0.541	0/20
BC23944_D2	0.976	0.507	0.558	0/20
BC23944_E1	0.980	0.498	0.533	0/20

Section	True AUC	Null median	Null p95	Exceedances
BC23944_E2	0.993	0.518	0.569	0/20
BC24044_D2	0.973	0.500	0.559	0/20
BC24044_E1	0.963	0.503	0.570	0/20
BC24044_E2	0.972	0.514	0.578	0/20
BC24105_D1	0.065	0.585	0.899	18/20
BC24220_D2	0.924	0.500	0.574	0/20
BC24220_E1	0.954	0.508	0.575	0/20
BC24220_E2	0.941	0.524	0.591	0/20

P Multi-task per-fold results

Table 21: Multi-task legs per held-out patient (median per-gene Pearson over the 250 train-selected genes). ridge = committed independent ridge; refold = same ridge refit by the multi-task harness (sanity); PLS = partial least squares (25 components); RRR = reduced-rank ridge (rank 25); enet = multi-output elastic net (alpha 1e-3, l1-ratio 0.5, max 500 iterations, random selection, seed 20260929; all folds hit the iteration cap - non-convergence declared).

patient	ridge	refold	PLS	RRR	enet	enet n_iter
BC23287	0.0588	0.0520	0.0896	0.0721	0.0467	500
BC23450	0.0388	0.0484	0.0734	0.0562	0.0439	500
BC23567	0.0334	0.0404	0.0491	0.0500	0.0386	500
BC23810	0.0623	0.0720	0.0786	0.0766	0.0658	500
BC23901	0.0417	0.0314	0.0494	0.0390	0.0277	500
BC23903	0.1063	0.1006	0.1077	0.1041	0.0948	500
BC23944	0.0390	0.0323	0.0442	0.0302	0.0268	500
BC24044	0.0853	0.0846	0.0966	0.0910	0.0797	500
BC24105	0.0312	0.0315	0.0400	0.0366	0.0279	500
BC24220	0.0507	0.0542	0.0611	0.0630	0.0511	500

Q Phase-2 per-fold results

Per-patient held-out summaries for the Phase-2 ridge probe, from `results/p2_folds_partial.csv`: per-gene medians (M1 ridge, M2 smoothed), the pathway-level median that carries the headline result, and the immune-composition AUC. λ and α are chosen by inner cross-validation on training patients only.

R G2-X v2 per-pathway external transfer

All 50 pathway-level Pearson correlations behind the locked G2-X v2 verdict, from `results/g2x_external_eval_v2.json` (4,325 external 10x Visium spots, frozen head, Reinhard-normalized tiles;

Table 22: Phase-2 per-LOPO-fold medians.

patient	λ	α	M1 gene r	M2 gene r	pathway r	immune AUC
BC23287	100	0.5	0.059	0.072	0.642	0.783
BC23450	100	0.5	0.039	0.042	0.246	0.573
BC23567	100	0.5	0.033	0.039	0.104	0.520
BC23810	100	0.5	0.062	0.068	0.101	0.577
BC23901	100	0.5	0.042	0.038	0.400	0.677
BC23903	100	0.5	0.106	0.117	0.614	0.743
BC23944	100	0.5	0.039	0.044	0.252	0.590
BC24044	100	0.5	0.085	0.098	0.571	0.793
BC24105	100	0.5	0.031	0.038	0.093	0.577
BC24220	100	0.5	0.051	0.061	0.497	0.696

gate median ≥ 0.10 FAILED at -0.0728). Sorted descending so the transfer gradient is inspectable: even the best-transferred pathway stays far below the gate.

pathway	r
Angiogenesis	+0.226
Apical Surface	+0.183
Epithelial Mesenchymal Transition	+0.138
Bile Acid Metabolism	+0.105
Glycolysis	+0.090
Spermatogenesis	+0.083
TGF-beta Signaling	+0.082
Coagulation	+0.064
KRAS Signaling Up	+0.029
Wnt-beta Catenin Signaling	+0.023
Myogenesis	+0.011
heme Metabolism	+0.002
Protein Secretion	-0.002
Pancreas Beta Cells	-0.002
Hypoxia	-0.014
UV Response Dn	-0.016
E2F Targets	-0.018
Inflammatory Response	-0.026
Myc Targets V1	-0.041
Complement	-0.053
IL-2/STAT5 Signaling	-0.054
Mitotic Spindle	-0.061
Reactive Oxygen Species Pathway	-0.062
Myc Targets V2	-0.068
mTORC1 Signaling	-0.072
Cholesterol Homeostasis	-0.074
Apical Junction	-0.088
KRAS Signaling Dn	-0.093
Oxidative Phosphorylation	-0.100
Hedgehog Signaling	-0.104
Androgen Response	-0.107

pathway	<i>r</i>
Xenobiotic Metabolism	-0.112
IL-6/JAK/STAT3 Signaling	-0.138
UV Response Up	-0.149
p53 Pathway	-0.158
TNF-alpha Signaling via NF-kB	-0.161
Apoptosis	-0.172
Peroxisome	-0.181
Notch Signaling	-0.183
Fatty Acid Metabolism	-0.183
Allograft Rejection	-0.191
G2-M Checkpoint	-0.194
Unfolded Protein Response	-0.195
Estrogen Response Early	-0.196
Adipogenesis	-0.196
DNA Repair	-0.267
Interferon Gamma Response	-0.317
Interferon Alpha Response	-0.326
Estrogen Response Late	-0.333
PI3K/AKT/mTOR Signaling	-0.382

Table 23: G2-X v2 per-pathway external-transfer correlations (sorted).

S M3 per-fold results

Per-patient held-out medians for the M3 MLP nonlinear check against the committed M1/M2 values (same folds, same features; M3 spec in Table 1). M3 loses to the ridge in 9 of 10 folds.

Table 24: M3 vs M1/M2 per LOPO fold (median per-gene Pearson).

patient	M1 ridge	M2 smoothed	M3 MLP	M3 – M1
BC23287	0.0598	0.0672	0.0047	-0.0551
BC23450	0.0427	0.0440	0.0177	-0.0250
BC23567	0.0214	0.0203	0.0196	-0.0019
BC23810	0.0286	0.0276	0.0320	+0.0035
BC23901	0.0093	0.0072	-0.0115	-0.0208
BC23903	0.0686	0.0749	0.0522	-0.0165
BC23944	0.0338	0.0375	0.0059	-0.0279
BC24044	0.0710	0.0766	0.0612	-0.0098
BC24105	-0.0016	-0.0093	-0.0172	-0.0156
BC24220	0.0458	0.0469	0.0257	-0.0201

Table 25: Track B corrected-geometry rerun (results/trackB_v2_folds_partial.csv): patient-held-out scanner performance on the corrected handcrafted features (features_cache2, 100% of spots in-bounds on all 68 sections), identical recipe to the original run (class-balanced logistic regression and histogram gradient boosting, LOPO over the same 23 patients). Median AUC: LR 0.8915, GB 0.8772 - both above the locked G-B1 gate of 0.80, superseding the buggy-geometry measurements (0.642 / 0.676). The four folds below 0.70 (BC23270, BC24105, BC24223, BC23567) sit in near-degenerate or near-balanced label strata and are named, not removed.

Patient	<i>n</i> spots	tumor frac	AUC (LR)	AUC (GB)	Brier (LR)
BC23209	943	0.950	0.942	0.932	0.052
BC23268	1425	0.182	0.795	0.813	0.134
BC23269	1302	0.845	0.972	0.963	0.131
BC23270	851	0.999	0.235	0.444	0.052
BC23272	1691	0.671	0.875	0.877	0.285
BC23277	1585	0.736	0.978	0.951	0.069
BC23287	947	0.276	0.827	0.875	0.142
BC23288	1422	0.907	0.808	0.767	0.135
BC23377	1885	0.756	0.939	0.949	0.132
BC23450	935	0.908	0.949	0.916	0.110
BC23506	1473	0.940	0.964	0.973	0.069
BC23508	1455	0.593	0.910	0.892	0.158
BC23567	1729	0.485	0.689	0.757	0.428
BC23803	1163	0.944	0.961	0.861	0.191
BC23810	2099	0.881	0.697	0.773	0.116
BC23895	1470	0.954	0.958	0.942	0.039
BC23901	589	0.486	0.754	0.593	0.203
BC23903	1373	0.199	0.892	0.878	0.106
BC23944	1240	0.836	0.968	0.966	0.195
BC24044	1728	0.123	0.918	0.932	0.113
BC24105	960	0.999	0.693	0.531	0.043
BC24220	1297	0.908	0.880	0.877	0.327
BC24223	1093	0.091	0.656	0.667	0.158
Median			0.8915	0.8772	

Table 26: Track B corrected-geometry scanner: feature-level inference. statsmodels logistic fit on standardized corrected features, all sections; coefficients sorted by $|z|$. Stain and color channels (blue-channel mean, red-channel mean, eosin and DAB deconvolution means) carry the largest effects, with GLCM texture statistics behind them - the scanner reads stain composition and local disorganization, exactly the morphology a pathologist weighs. Supersedes the buggy-geometry inference (results/scanner_coefficients.csv, preserved as audit trail).

Feature	Coef	z	p
b_mean	+9.659	+17.30	4.54e-67
r_mean	-5.694	-17.84	3.63e-71
eosin_mean	+5.144	+12.06	1.69e-33
dab_mean	+4.933	+13.69	1.24e-42
glcm3_contrast	+4.388	+3.85	1.20e-04
glcm1_energy	-4.344	-2.72	6.46e-03
glcm3_corr	+4.009	+5.54	3.03e-08
glcm1_contrast	-3.606	-3.32	9.15e-04
glcm1_corr	-3.581	-4.95	7.29e-07
r_kurt	-3.573	-5.68	1.32e-08
glcm2_homog	-3.199	-5.48	4.21e-08
hema_mean	+2.649	+9.74	1.94e-22
glcm0_energy	+2.168	+1.38	1.68e-01
r_std	+2.140	+11.38	5.33e-30
b_skew	+2.139	+11.62	3.25e-31
glcm0_homog	+1.951	+3.31	9.27e-04
glcm2_contrast	+1.940	+1.68	9.34e-02
b_kurt	+1.906	+6.72	1.76e-11
hema_std	-1.750	-14.52	9.56e-48
eosin_std	+1.539	+8.17	3.18e-16
glcm2_corr	+1.431	+1.95	5.10e-02
g_std	-1.431	-6.72	1.77e-11
gray_entropy	-1.385	-16.11	2.15e-58
glcm0_contrast	-1.309	-1.19	2.36e-01
b_std	-1.292	-8.54	1.29e-17
glcm0_corr	-1.232	-1.69	9.10e-02
glcm2_energy	+1.212	+0.97	3.32e-01
g_mean	-1.150	-3.38	7.21e-04
hema_dense_frac	+0.952	+3.44	5.86e-04
eosin_fracpos	-0.764	-11.22	3.33e-29
glcm3_energy	+0.731	+0.57	5.66e-01
glcm1_homog	+0.603	+1.03	3.03e-01
g_kurt	-0.541	-1.20	2.29e-01
glcm3_homog	-0.538	-0.94	3.48e-01

Feature	Coef	z	p
hema_fracpos	-0.435	-5.39	6.97e-08
grad_mag_mean	+0.397	+2.19	2.82e-02
g_skew	-0.300	-1.32	1.87e-01
hema_entropy	+0.205	+5.09	3.52e-07
r_skew	+0.155	+0.66	5.08e-01
dab_fracpos	+0.131	+2.72	6.50e-03
dab_std	+0.129	+2.00	4.52e-02

Table 27: Learned versus handcrafted representation, per held-out patient. Handcrafted: Track A M1 ridge on 41 morphology features (corrected-geometry rerun); learned: Phase-2 ridge on 768-dimensional CTransPath embeddings. Values are median per-gene Pearson r on the held-out patient’s spots. The learned representation wins 8 of 10 folds (median paired delta +0.0131, Wilcoxon $p = 0.0098$); both medians remain far below the locked G2-R gate of 0.08.

Patient	Handcrafted r	Learned r	Δ (learned – handcrafted)
BC23287	0.0598	0.0588	-0.0010
BC23450	0.0427	0.0388	-0.0039
BC23567	0.0214	0.0334	+0.0120
BC23810	0.0286	0.0623	+0.0337
BC23901	0.0093	0.0417	+0.0324
BC23903	0.0686	0.1063	+0.0377
BC23944	0.0338	0.0390	+0.0052
BC24044	0.0710	0.0853	+0.0142
BC24105	-0.0016	0.0312	+0.0328
BC24220	0.0458	0.0507	+0.0049
Median	0.0382	0.0462	+0.0131

Table 28: Nested evaluation of the 250-gene selection rule (pre-declared rules, queue #11). Per-fold median per-gene Pearson r on the held-out patient for four selection rules: top-100 / top-250 / top-500 by train-fold mean expression, and top-250 by train-fold variance. The last column is the gene overlap between the mean- and variance-selected 250-gene panels. The headline median moves by at most ± 0.005 across rules and every rule remains far below the locked G2-R gate of 0.08 - the gate conclusion is selection-rule-robust.

Patient	top-100 mean	top-250 mean	top-500 mean	top-250 var	mean $\cap$ var
BC23287	0.0366	0.0520	0.0775	0.0767	155

Patient	top-100 mean	top-250 mean	top-500 mean	top-250 var	mean $\cap$ var
BC23450	0.0836	0.0484	0.0473	0.0545	172
BC23567	0.0420	0.0404	0.0357	0.0392	168
BC23810	0.1121	0.0720	0.0471	0.0592	176
BC23901	0.0402	0.0314	0.0353	0.0303	172
BC23903	0.1197	0.1006	0.0894	0.0958	171
BC23944	0.0338	0.0323	0.0321	0.0326	177
BC24044	0.0765	0.0846	0.0889	0.0855	161
BC24105	0.0509	0.0315	0.0251	0.0356	170
BC24220	0.0633	0.0542	0.0492	0.0547	174
Median	0.0571	0.0502	0.0472	0.0546	171.5

Table 29: Gene-level negative control (pre-declared, queue #12). Control column: median across 5 pre-declared draws of 250 genes from train-expression ranks 251-2250 (seed 20260929), same ridge harness. Selected-250: the committed ridge on the fold-varying top-250 panel. PLS: the multi-task winner. The control's pre-declared ≈ 0 expectation was falsified - per-gene predictability is broad across expressed genes (cell-type-composition-driven), not panel-specific - but the selected panel beats controls in 9/10 folds (Wilcoxon $p = 0.0488$) and PLS in 10/10 ($p = 0.0020$). Named anomaly: BC23287, controls $0.0814 > \text{selected } 0.0588$.

Patient	Control median r	Selected-250 ridge r	PLS r
BC23287	0.0806	0.0588	0.0896
BC23450	0.0303	0.0388	0.0734
BC23567	0.0223	0.0334	0.0491
BC23810	0.0220	0.0623	0.0786
BC23901	0.0365	0.0417	0.0494
BC23903	0.0654	0.1063	0.1077
BC23944	0.0298	0.0390	0.0442
BC24044	0.0865	0.0853	0.0966
BC24105	0.0112	0.0312	0.0400
BC24220	0.0390	0.0507	0.0611
Median	0.0334	0.0462	0.0673

Table 30: Spatial structure of the expression task, per held-out patient. kNN / Gaussian oracles: median per-gene r of smoothing predictors given the *true* expression at neighboring held-out spots (an upper bound on what pure spatial autocorrelation carries). Moran’s I : median spatial autocorrelation of the 250-gene panel; its correlation with the kNN-oracle r shows per-gene predictability tracks autocorrelation. Gain: M2 (spatial-feature model) minus M1 held-out r ; shell p95: 95th percentile of the same gain under distance-shell permutations (the strictest of three pre-declared nulls).

Patient	kNN oracle	Gaussian oracle	Moran’s I	corr(I , kNN r)	M2–M1 gain	shell null p95
BC23287	0.123	0.146	0.029	0.89	+0.0138	0.0173
BC23450	0.110	0.137	0.038	0.94	-0.0001	0.0070
BC23567	0.166	0.214	0.071	0.98	+0.0074	0.0099
BC23810	0.112	0.151	0.045	0.96	+0.0053	0.0024
BC23901	0.114	0.141	0.038	0.95	-0.0041	0.0033
BC23903	0.158	0.184	0.053	0.96	+0.0114	0.0096
BC23944	0.108	0.143	0.042	0.95	+0.0075	0.0165
BC24044	0.165	0.201	0.062	0.97	+0.0133	0.0095
BC24105	0.105	0.116	0.034	0.96	+0.0067	0.0072
BC24220	0.090	0.114	0.033	0.97	+0.0071	0.0091
Median	0.113	0.144	0.040	0.96	+0.0073	0.0093

T Scanner calibration reliability table

The full 10-bin reliability table behind the calibration diagnostics, from `results/p2_scanner_calibration.json` (12,897 out-of-fold spots; Brier 0.172, prevalence baseline 0.240, ECE 0.141).

Table 31: Scanner reliability by predicted-probability bin.

bin	n	mean predicted	empirical tumor rate	gap
0.0–0.1	3,588	0.015	0.140	-0.124
0.1–0.2	471	0.146	0.433	-0.288
0.2–0.3	372	0.249	0.578	-0.329
0.3–0.4	322	0.350	0.643	-0.293
0.4–0.5	327	0.449	0.667	-0.217
0.5–0.6	335	0.547	0.716	-0.170
0.6–0.7	441	0.650	0.755	-0.106
0.7–0.8	536	0.753	0.750	+0.003
0.8–0.9	855	0.856	0.794	+0.062
0.9–1.0	5,650	0.979	0.840	+0.140

U Transport cohort, all 39 sections

Every external-transport measurement behind G-A4, from `results/trackA_results.json`: the HER2+-trained M1 ridge applied to Luminal A, Luminal B, and TNBC sections with no refit. Sorted by subtype

then correlation; the degradation versus in-subtype held-out performance is the expected, honestly-logged negative.

section	subtype	median r
Luminal_A (n=12, median 0.070)		
BC23268_D1	Luminal_A	0.1311
BC23268_C2	Luminal_A	0.1124
BC23268_C1	Luminal_A	0.1122
BC24223_E1	Luminal_A	0.0872
BC24223_D2	Luminal_A	0.0864
BC23272_E1	Luminal_A	0.0697
BC23272_D2	Luminal_A	0.0565
BC23269_C1	Luminal_A	0.0450
BC24223_E2	Luminal_A	0.0438
BC23272_E2	Luminal_A	0.0374
BC23269_D1	Luminal_A	0.0318
BC23269_C2	Luminal_A	0.0186
Luminal_B (n=15, median 0.027)		
BC23508_D2	Luminal_B	0.0578
BC23277_E1	Luminal_B	0.0557
BC23270_E1	Luminal_B	0.0505
BC23508_E1	Luminal_B	0.0453
BC23508_E2	Luminal_B	0.0373
BC23277_D2	Luminal_B	0.0323
BC23895_C1	Luminal_B	0.0288
BC23895_D1	Luminal_B	0.0273
BC23506_C1	Luminal_B	0.0244
BC23506_C2	Luminal_B	0.0026
BC23895_C2	Luminal_B	0.0007
BC23270_D2	Luminal_B	-0.0032
BC23270_E2	Luminal_B	-0.0032
BC23506_D1	Luminal_B	-0.0045
BC23277_E2	Luminal_B	-0.0133
TNBC (n=12, median 0.029)		
BC23209_D1	TNBC	0.0711
BC23209_C2	TNBC	0.0553
BC23288_E2	TNBC	0.0536
BC23288_E1	TNBC	0.0437
BC23803_E1	TNBC	0.0385
BC23288_D2	TNBC	0.0286
BC23209_C1	TNBC	0.0188
BC23803_D2	TNBC	0.0049
BC23377_C1	TNBC	-0.0088
BC23803_E2	TNBC	-0.0186
BC23377_D1	TNBC	-0.0277
BC23377_C2	TNBC	-0.0348

section	subtype	median r
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Table 32: G-A4 transport: per-section held-out median per-gene Pearson on non-HER2 subtypes.

V Per-pathway niche spatial architecture (D2)

All 50 pathway-niche Moran's I values behind the locked D2 result, from `results/p2_niches_summary.json` ($k = 8$ spatial niches per section, 200 null relabelings; 29 HER2 sections). Sorted by median I across sections; $\text{frac } p < 0.05$ is the fraction of sections where the niche's pathway score is significantly spatially clustered. Cohort medians: niche I 0.213; 96.6% of sections have all niches significant; the PancreasBeta control sits at 0.072.

pathway niche	median I	frac $p < 0.05$
Myc Targets V1	0.362	1.00
Mitotic Spindle	0.346	1.00
G2-M Checkpoint	0.340	1.00
mTORC1 Signaling	0.339	1.00
E2F Targets	0.329	1.00
Glycolysis	0.308	1.00
Epithelial Mesenchymal Transition	0.305	1.00
Androgen Response	0.303	1.00
Oxidative Phosphorylation	0.302	1.00
Unfolded Protein Response	0.301	1.00
Interferon Gamma Response	0.288	1.00
Xenobiotic Metabolism	0.287	1.00
Protein Secretion	0.285	1.00
Fatty Acid Metabolism	0.278	1.00
Hypoxia	0.277	1.00
Adipogenesis	0.275	1.00
DNA Repair	0.274	1.00
Apical Junction	0.273	1.00
Estrogen Response Early	0.270	1.00
Cholesterol Homeostasis	0.265	1.00
TNF-alpha Signaling via NF-kB	0.262	1.00
heme Metabolism	0.259	1.00
UV Response Up	0.259	1.00
Estrogen Response Late	0.258	1.00
Pperoxisome	0.256	1.00
p53 Pathway	0.255	0.97
UV Response Dn	0.255	1.00
Complement	0.252	1.00
Apoptosis	0.247	0.97
PI3K/AKT/mTOR Signaling	0.247	1.00
IL-2/STAT5 Signaling	0.238	1.00
Interferon Alpha Response	0.232	1.00

pathway niche	median I	frac $p < 0.05$
Myogenesis	0.231	1.00
Coagulation	0.229	1.00
KRAS Signaling Up	0.222	0.93
Allograft Rejection	0.202	1.00
TGF-beta Signaling	0.199	0.93
Inflammatory Response	0.195	0.97
Angiogenesis	0.184	0.90
Myc Targets V2	0.183	0.93
Bile Acid Metabolism	0.165	0.93
Reactive Oxygen Species Pathway	0.164	0.97
Spermatogenesis	0.163	0.93
IL-6/JAK/STAT3 Signaling	0.144	0.97
Notch Signaling	0.142	0.90
Apical Surface	0.137	0.93
Wnt-beta Catenin Signaling	0.125	0.83
KRAS Signaling Dn	0.114	0.90
Hedgehog Signaling	0.105	0.83
Pancreas Beta Cells	0.072	0.83

Table 33: Per-pathway niche spatial clustering (sorted by median Moran's I).